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Failure-specific cooperative recourse strategy for simultaneous pickup and delivery problem with stochastic demands

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ABSTRACT

This paper concerns the generation of a priori routes for a fleet of vehicles that pick up and deliver items with stochastic demands. A failure-specific cooperative recourse strategy is proposed to explore a risk pooling mechanism for routing in the context of simultaneous pickup and delivery with stochastic demands. By defining complete failure and semi-failure of routing, the travelling cost under our failure-specific cooperative strategy is estimated. Also, an adaptive large neighbourhood search algorithm is developed. Compared with a strategy that involves no cooperation between vehicles, our strategy performs better in terms of reducing travelling costs, and balancing fleet size and detour frequency.

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1. Introduction

The vehicle routing problem with simultaneous pickup and delivery (VRPSPD) is an extension of the classic vehicle routing problem (VRP). In the VRPSPD, a vehicle departs from the depot, picks up items from and simultaneously delivers goods to customers, and finally returns to the depot (Berbeglia, Cordeau, Gribkovskaia, & Laporte, 2007; Berbeglia, Cordeau, & Laporte, 2010; Polat, Kalayci, Kulak, & Günther, 2015). The picked up and delivered items are regarded as two types of products: the picked up items are referred to as returned (recycled or retired), while the delivered items are new products. This problem is also relevant to reverse logistics (Dell'Amico, Righini, & Salani, 2006; Nagy, Wassan, Speranza, & Archetti, 2015).

This paper studies the VRPSPD with stochastic demands (SD), denoted *simultaneous pickup and delivery problem with stochastic demands* (SPDPSPD), referring to the situation where customer demand for pickup or delivery is stochastic (or both). Under the SPDPSPD, a vehicle may reach a customer location without enough capacity to pick up returned items or sufficient inventory to deliver new products (or both). This causes a route *failure* that requires a *recourse* action. Various recourse actions are possible (Zhu, Rousseau, Rei, & Li, 2014): (i) replenishing the vehicle at the depot, (ii) scheduling another vehicle to visit the customer where the failure occurred, and (iii) skipping the customer altogether (a

penalty is incurred in this case). We consider (i) and (ii), situations in which a driver performs a detour trip to the depot or leaves the demand to be satisfied by another driver.

Research relevant to the SPDPSPD is scant when considering restocking policies (e.g., Dimitrakos & Kyriakidis, 2015; Minis & Tatarakis, 2011; Pandelis, Karamatsoukis, & Kyriakidis, 2013; Pandelis, Kyriakidis, & Dimitrakos, 2012) or routing decisions (e.g., Hou & Zhou, 2010; Hu, Sheu, Zhao, & Lu, 2015), and few works pay attention to the construction of a priori routes. To the best of our knowledge, only Wollenberg (2015) observed this gap and studied a situation where delivery demand was known with certainty. We deal with the SPDPSPD when all customer demands are stochastic, and we generate a priori routes for simultaneous pickup and delivery with uncertain demands.

To obtain a priori routes, stochastic programming with recourse (SPR) is often used. The travelling cost of a route is approximated by the total of the cost of the planned route and the cost of expected recourse actions (e.g., Laporte, Louveaux, & Van Hamme, 2002). Ak and Erera (2007) proposed a recourse strategy, named paired locally-coordinated (PLC), which followed the SPR standard and embedded locally-coordinated mechanism. PLC reduces the travelling cost through vehicle cooperation. Using the PLC, each vehicle in a pair serves customers sequentially, following a fixed route. If one vehicle fails, it ceases its service immediately and all unserved customers are assigned to its partner (see Ak & Erera, 2007). But this strategy is not applicable to the problem that we concern. In the SPDPSPD, route failure may be frequent (due to insufficient capacity for a pickup or inventory for a delivery). If the PLC strategy is followed, frequent and undesired vehicle interac-

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tion may occur; a vehicle may fail and cease service at the first customer and put all remaining burdens on its partner.

We propose a cooperative recourse strategy that is applicable to the SPDPSPD. In our strategy, if a vehicle fails, it resumes its route to serve its remaining assigned customers, rather than leaving all of them to its partner.

We identify two kinds of failure for a vehicle serving customers with pickup and delivery demand. The first type of failure is called *complete-failure* (CF or C-failure), and it refers to the case where a vehicle is fully loaded with returned items; in this case, the vehicle, with neither capacity for a pickup nor new products for delivery, has to unload and reload at the depot before resuming its route. The second type of failure is *semi-failure* (SF or S-failure); in this case, there is still either sufficient capacity for a pickup at or inventory for delivery to another customer, and the vehicle need not return to the depot.

The development of our *failure-specific cooperative recourse* (FSC) strategy is based on these two types of failure. Each pair of vehicles is composed of a lead vehicle and a partner vehicle; the lead vehicle follows a fixed route to serve its customers, and the partner vehicle is scheduled to assist the lead vehicle in completing its service while also serving its own assigned customers. The benefit of letting the partner vehicle take over is greatest, when the depot is distant and the partner vehicle is near the lead vehicle.

The probabilities of C-failure and S-failure are used in computing the total expected travelling cost. Given recourse actions in FSC strategy, the probability of C-failure is used to compute the expected recourse cost of the lead vehicle, and the probability of S-failure is relevant to the possibility of the partner vehicle visiting a customer in order to satisfy the customer's remaining demand.

This paper's main contribution is the development of an FSC strategy to build a priori routes for simultaneous pickup and delivery. When a C-failure or S-failure is encountered, this strategy informs the lead vehicle to satisfy unmet demand after returning to the depot or leave unmet demand to its partner vehicle. We compute the expected travelling cost of each vehicle that follows an a priori route and takes recourse actions according to FSC strategy. We develop an adaptive large neighbourhood search (ALNS) algorithm that optimises routing sequences. We compare the performance of FSC strategy (through a numerical study that shows its benefit in terms of both real travelling costs and in the balance of fleet size and detour frequency) to that of another recourse strategy.

The remainder of this paper is organised as follows: [Section 2](#) describes the problem we concern; [Section 3](#) illustrates FSC strategy; [Section 4](#) estimates the expected cost of a pair of routes; [Section 5](#) presents the ALNS algorithm; and [Section 6](#) compares FSC strategy with another recourse strategy and discusses relevant numerical results.

2. Problem description and notations

Consider a complete network, and let the set of nodes be $\{0, 1, \dots, N\}$, where N is a positive integer. Node 0 denotes the depot and $C = \{1, \dots, N\}$ is the set of customers. The travelling distance $d(i, j)$ between any nodes i and j is symmetric and satisfies the triangular inequality. A fleet of vehicles with identical capacity Q operates to meet pickup and delivery demands. The number of vehicles in use is determined by the operator, with sufficient vehicles in the fleet. Assume that customer demands are identically distributed, and ξ_i and φ_i ($i = 1, \dots, N$) are the discrete random variables that describe the amount of delivered and picked up items demanded by customer i . The probability mass functions are $q(e) = \Pr\{\xi_i = e\}$, $e = 0, 1, \dots, Q$ and $h(t) = \Pr\{\varphi_i = t\}$, $t = 0, 1, \dots, Q$, respectively, where demands ξ_i and φ_i are mu-

tually independent and observed only when a vehicle arrives. Each vehicle starts from the depot, serves its assigned customers following a predefined routing sequence, and eventually returns to the depot. The problem is to design the a priori route for each vehicle in use, and to reduce the total travelling cost.

Assume that a vehicle arrives at new customer i , it first delivers as many new products as possible to satisfy demand ξ_i , and then picks up the largest possible quantity of returned items to meet demand φ_i . After the first visit of customer i , the vehicle's inventory of new products is reduced to z_i , and its available capacity changes to r_i . z_i can be less than 0, indicating the quantity of new products still needed to complete delivery service to customer i , and $z_i^- = -\min\{0, z_i\}$ denotes the amount owed to be delivered. r_i can also be less than 0, indicating the quantity of returned items from customer i that cannot be loaded on the vehicle, and $r_i^- = -\min\{0, r_i\}$ denotes the amount owed to be picked up; when r_i is positive, it refers to the empty space that has not been occupied by any items, either returned or new products (i.e., $r_i = Q - \max\{z_i, 0\} - \{\text{amount of returned items after loading}\}$, if $r_i > 0$).

Let (z_i, r_i) denote the state of a vehicle after the first visit of customer i and the time that is used as the decision epoch for scheduling a vehicle to visit a new customer, either directly or via a detour trip to the depot. Ψ_i denotes the state set after the vehicle's first visit of customer i , i.e., $\forall (z_i, r_i) \in \Psi_i$. The cardinality of Ψ_i is $(2Q + 1)^2 - \frac{1}{2}Q(Q + 1)$, by enumerating the values of (z_i, r_i) when requirements $z_i \in \{-Q, \dots, Q\}$, $r_i \in \{-Q, \dots, Q\}$ and $z_i + r_i \leq Q$ are satisfied. θ_i refers to the quantity of new products to be restocked at the depot after the vehicle completes its service of customer i .

Assume that a customer can be served by more than one vehicle. If a failure occurs at customer i , the vehicle should satisfy as much of the delivery and pickup demand of customer i as possible, before leaving the remainder of the demand to be satisfied by itself on a return trip or by another vehicle.

3. Failure-specific cooperative recourse strategy for the SPDPSPD

FSC is a recourse strategy that allows cooperation between each pair of vehicles. A subset of customers is assigned to a pair of vehicles, and each customer is served by one or both of the vehicles. In each pair, the lead vehicle follows a type A route, serving each customer in turn; in the event of an S-failure, it proceeds to the next customer, and in the event of a C-failure it returns to the depot to unload and reload. Meanwhile, the partner vehicle follows a type B route, and meets any demand not met by the lead vehicle.

Let C_p ($\bigcup_p C_p = C$) be the subset of customers that a pair of vehicles, V_A^p and V_B^p serve, where V_A^p and V_B^p follow type A and type B routes respectively, simply denoted by V_A and V_B . One example of a pair of vehicles travelling under FSC strategy is shown in [Fig. 1](#).

[Fig. 1\(a\)](#) shows the a priori routes for a pair of vehicles. [Fig. 1\(b\)](#) shows that vehicle V_A follows a type A route, with C-failures at customer sites 2 and 8 and S-failures at customer sites 5 and 7; the vehicle proceeds to the next customer when an S-failure occurs and makes a detour trip to the depot when a C-failure occurs. In [Fig. 1\(c\)](#), customers 5 and 7 with unmet demands are revisited by the partner vehicle by adding those customers to the a priori type B route; the partner vehicle then meets the remaining customer demands altogether, making detour trips immediately after each subsequent failure, regardless of type.

Under FSC strategy, vehicle V_A , following a type A route, reacts differently depending on encountering a C-failure or an S-failure. With a C-failure, the vehicle is full of returned items and has to detour to the depot before resuming its route. Under this condition, the vehicle not only fails to meet the delivery demand of customer i (ξ_i), but also fails to satisfy the pickup demand of customer i (φ_i),

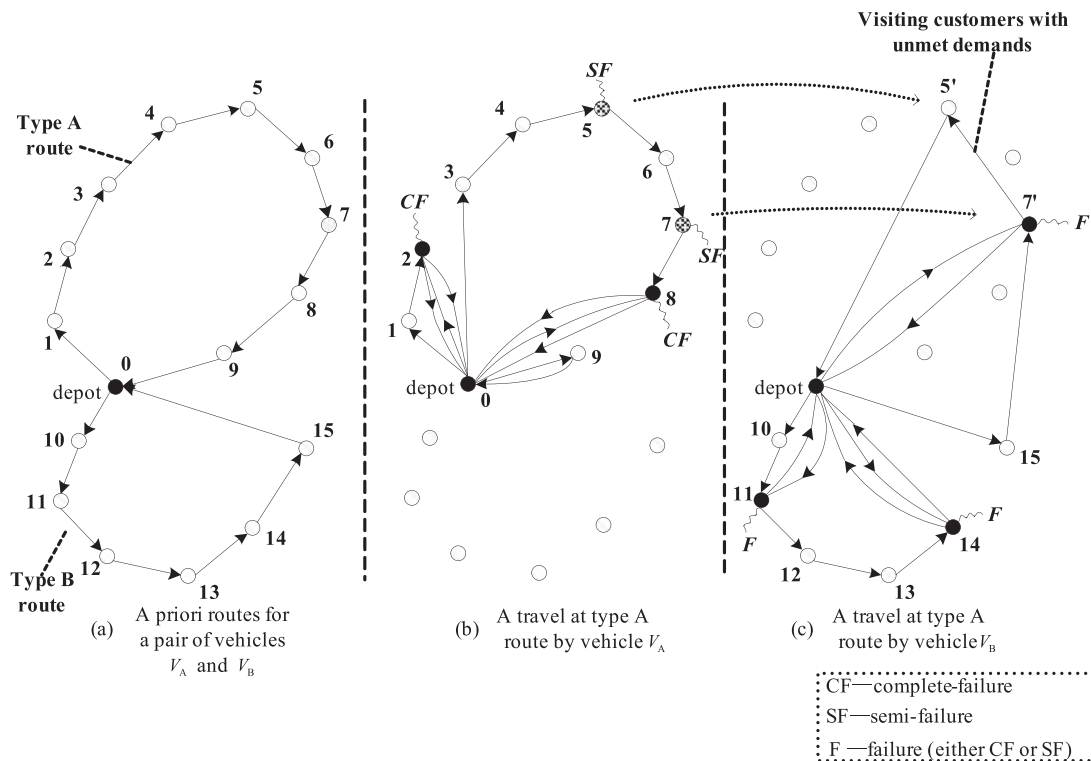


Fig. 1. Example of a pair of vehicles travelling under FSC strategy.

carries no remaining new products ($z_i \leq 0$), and has no further capacity for picking up returned items ($r_i \leq 0$). This makes it impossible for it to serve any additional customer, which is denoted by ($z_i \leq 0, r_i \leq 0$). However, with an S-failure, the vehicle either fails to meet the delivery demand of customer i or fails to satisfy the pickup demand of customer i – but it still has the potential to serve additional customers. The S-failure refers to two conditions: (i) available inventory to deliver further new products but the lack of capacity for further item pickups, indicated by ($z_i > 0, r_i \leq 0$); (ii) available capacity for further pickups but a lack of inventory of new products to deliver, indicated by ($z_i \leq 0, r_i > 0$).

In the next section, we calculate the expected travelling cost of a pair of vehicles V_A and V_B . The two vehicles serve a shared customer subset C_p , and take recourse actions based on FSC strategy.

4. Expected cost of one pair of routes

Let customers in subset C_p ($C_p \subseteq C$) be covered by a pair of a priori routes τ_A and τ_B , type A and B routes respectively, requiring that $C^{\tau_A} \cap C^{\tau_B} = \emptyset$ and $C^{\tau_A} \cup C^{\tau_B} = C_p$, where C^{τ} refers to the customers who are covered by route τ . Vehicles V_A and V_B are scheduled to follow two routes τ_A and τ_B , and eventually travel as shown in Fig. 1(b) and (c). Section 4.1 shows the expected travelling cost of vehicle V_A , denoted by $G(V_A)$, and Section 4.2 shows the expected travelling cost of vehicle V_B , $G(V_B)$.

4.1. Expected cost of a type A route

Vehicle V_A only makes a detour trip to the depot when it encounters a C-failure at a customer site. The travelling cost is the length of the planned route and the expected distance of recourse actions; that is, $G(V_A) = \text{Len}(\tau_A) + \sum_{i \in C^{\tau_A}} \mathcal{P}^{CF}(i) \cdot D^{\text{recs}}(i)$, where $\mathcal{P}^{CF}(i)$ is the probability that a C-failure occurs at customer site i ($i \in C^{\tau_A}$), and $D^{\text{recs}}(i)$ is the distance travelled when vehicle V_A makes a detour trip to the depot from customer i .

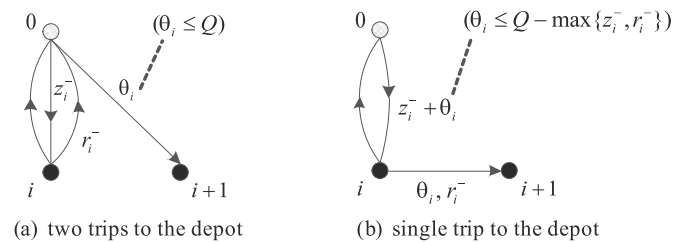


Fig. 2. Two kinds of recourse action taken in simultaneous pickup and delivery problems.

Assume that Vehicle V_A follows routing sequence $\tau_A = [0, 1, \dots, i-1, i, i+1, \dots, n_A, 0]$ to serve customers in set C^{τ_A} ($|C^{\tau_A}| = n_A$), where i refers to the i th customer of the routing sequence. Section 4.1.1 introduces the recourse action taken by vehicle V_A when a C-failure occurs at a customer site, and thus obtains the recourse distance $D^{\text{recs}}(i)$ for each customer i ($i \in C^{\tau_A}$). Section 4.1.2 computes the probability $\mathcal{P}^{CF}(i)$ ($\forall i \in C^{\tau_A}$), and it is shown that we should obtain $\mathcal{P}^{CF(1)}(i)$ ($\forall i \in C^{\tau_A}$) first, which is the probability that a vehicle travels to customer site i where a C-failure occurs for the first time. Section 4.1.3 shows the details of calculating $\mathcal{P}^{CF(1)}(i)$, and Section 4.1.4 summarises the expected cost of vehicle V_A under FSC strategy.

4.1.1. Recourse action for a vehicle that follows a type A route

When a failure occurs at a customer site, the vehicle can complete serving the customer by one of two recourse actions: it can make two detour trips to the depot or just a single trip. This is shown in Fig. 2 (See Pandelis et al., 2013).

Fig. 2 shows the condition where a vehicle fails at customer site i ; that is, quantity z_i^- of new products is not delivered to and amount r_i^- of returned items is not picked up from customer i on the first visit. Fig. 2(a) indicates that a vehicle meets the remaining demand of customer i with two trips to the depot. After the vehi-

cle empties all items at the first visit of the depot, it only takes z_i^- new products back to the customer, collects the remaining returned items r_i^- , and travels to the depot for the second time. During the second trip, it unloads returned items and loads θ_i ($\theta_i \leq Q$) new products, then proceeds directly to the next customer.

Fig. 2(b) shows that a vehicle meets the remaining demand of customer i with a single trip to the depot. The vehicle travels to the depot, unloads all returned items and loads $z_i^- + \theta_i$ new products; then, it revisits customer i and meets its unmet demand z_i^- . The vehicle then proceeds to the next customer with some of its capacity occupied by returned items r_i^- . In the second recourse action, with only a single trip to the depot, θ_i is additionally limited by $Q - \max\{z_i^-, r_i^-\}$; the vehicle carries new products back to customer i , requiring that $\theta_i \leq Q - z_i^-$, and the vehicle picks up returned items from customer i , requiring that $\theta_i \leq Q - r_i^-$.

In the first recourse action, as shown in Fig. 2(a), after making two trips to the depot the vehicle proceeds to the next customer, carrying only new products for delivery. This state, where the only items it carries are new products, is similar to its initial state when it left the depot to serve the first customer and only carried items for delivery.

We manage that vehicle V_A makes two trips to the depot to unload and reload when it encounters a C-failure at a customer site. By fixing all recourse actions to two trips to the depot, the operation for vehicle V_A is simplified and computational effort is reduced. Therefore, the additional travelling cost of vehicle V_A taking a detour to the depot when encountering a C-failure at a customer site i is

$$D^{recs}(i) = 3d(i, 0) + d(0, i+1) - d(i, i+1), \quad i \in C^{\tau_A}. \quad (1)$$

In this equation, $3d(i, 0) + d(0, i+1)$ refers to the distance taken by two trips to the depot, as shown in Fig. 2(a), and $d(i, i+1)$ is part of the fixed route travelled if the vehicle does not encounter a C-failure at customer site i . The recourse cost $D^{recs}(i)$ is the difference between them, and the additional travelling cost is realised depending on the probability of a C-failure at the customer, $\mathcal{P}^{CF}(i)$.

4.1.2. Probability of C-failure at a customer site

Given FSC strategy, vehicle V_A may experience more than one C-failure when following route τ_A , and the vehicle does not make any recourse trips when travelling between the customer sites where two adjacent C-failures occur. (In Fig. 1(b), vehicle V_A encounters one C-failure at customer site 2 and a second C-failure at customer site 8, and no recourse trip is made by the vehicle between customer site 2 and 8.) Therefore, the probability of the k th ($k \geq 1$) C-failure at customer site i , denoted by $\mathcal{P}^{CF(k)}(i)$, depends on the probability of the $(k-1)$ th C-failure at a preceding customer site l ($l \leq i-1$), denoted by $\mathcal{P}^{CF(k-1)}(l)$, and the probability that no C-failure occurs between customer l and customer $i-1$, (in another word, the vehicle meets the first C-failure at customer i , from the last replenishment at customer l) denoted by $\mathcal{P}^{CF(1)}(i|l)$. This yields the relation

$$\mathcal{P}^{CF(k)}(i) = \sum_{l=k-1}^{i-1} \mathcal{P}^{CF(k-1)}(l) \cdot \mathcal{P}^{CF(1)}(i|l) \quad (2)$$

when following routing sequence $\tau_A = [0, 1, \dots, i-1, i, i+1, \dots, n_A, 0]$, vehicle V_A may experience the k th C-failure at customer site i , and k can take at most value i when the worst situation happens; that is, the vehicle encounters a C-failure at every customer site. For the $(k-1)$ th C-failure, and for the same reason, $k-1$ can take at most l . So, $k \leq i$, $k-1 \leq l$, and $k-1 \leq l \leq i-1$. Moreover, a vehicle may encounter the first, second, ..., i th C-failure at customer site i . The probability of

encountering a C-failure at customer site i can be expressed as

$$\mathcal{P}^{CF}(i) = \sum_{k=1}^i \mathcal{P}^{CF(k)}(i), \quad 1 \leq i \leq n_A. \quad (3)$$

Substituting Eq. (2) into Eq. (3), the probability of a C-failure at customer site i is

$$\mathcal{P}^{CF}(i) = \sum_{k=1}^i \sum_{l=k-1}^{i-1} \mathcal{P}^{CF(k-1)}(l) \cdot \mathcal{P}^{CF(1)}(i|l), \quad 1 \leq i \leq n_A. \quad (4)$$

Vehicle V_A takes two trips to the depot when a C-failure occurs at customer site l , and then encounters its first subsequent C-failure at customer site i . As explained above, the journey from customer site l to customer site i is similar to the journey in which the vehicle initially departs from the depot 0 until it meets the first failure at customer site $i-l$, yielding the relation

$$\mathcal{P}^{CF(1)}(i|l) = \mathcal{P}^{CF(1)}(i-l|0), \quad l \leq i-1. \quad (5)$$

From Eq. (2), we see that $\mathcal{P}^{CF(1)}(i) = \mathcal{P}^{CF(1)}(i|0)$, since $\mathcal{P}^{CF(1)}(i) = \mathcal{P}^{CF(0)} \cdot \mathcal{P}^{CF(1)}(i|0) + \sum_{l=1}^{i-1} \mathcal{P}^{CF(0)}(l) \cdot \mathcal{P}^{CF(1)}(i-l|0)$, and $\mathcal{P}^{CF(0)}(0) = 1$, $\mathcal{P}^{CF(0)}(i) = 0$ ($i \neq 0$) (meaning that the vehicle can only depart from the depot). Therefore, with Eqs. (4) and (5), we obtain the following equation for the probability of C-failure at customer site i :

$$\mathcal{P}^{CF}(i) = \sum_{k=1}^i \sum_{l=k-1}^{i-1} \mathcal{P}^{CF(k-1)}(l) \cdot \mathcal{P}^{CF(1)}(i-l), \quad 1 \leq i \leq n_A. \quad (6)$$

From Eqs. (3) and (6), in fact, $\mathcal{P}^{CF(k)}(i) = \sum_{l=k-1}^{i-1} \mathcal{P}^{CF(k-1)}(l) \cdot \mathcal{P}^{CF(1)}(i-l)$ holds, implying that a recursion relation exists for deducing the probability of the k th C-failure at a customer site, $\mathcal{P}^{CF(k)}(\cdot)$, given the probability of the $(k-1)$ th C-failure at each possible customer site l , $\mathcal{P}^{CF(k-1)}(\cdot)$. Therefore, if the first C-failure at each customer site i ($i = 1, 2, \dots, n_A$) is given, then the k th ($k \geq 2$) C-failure at a customer site can be computed, and finally the probability of a C-failure at each customer site i ($i \in C^{\tau_A}$) can be obtained from Eq. (6). The steps to obtain $\mathcal{P}^{CF}(i)$ follow

Algorithm 1 Steps to obtain $\mathcal{P}^{CF}(i)$.

- 1: **Step 1:** Compute the probability of the 1st C-failure at each customer site i , $\mathcal{P}^{CF(1)}(i)$ ($1 \leq i \leq n_A$),
- 2: **Step 2:** Compute the probability of the k th C-failure at each customer site i , $\mathcal{P}^{CF(k)}(i)$ ($k \geq 2$)
- 3: **for** $2 \leq k \leq n_A$
- 4: **for** $k \leq i \leq n_A$
- 5: Compute $\mathcal{P}^{CF(k)}(i) = \sum_{l=k-1}^{i-1} \mathcal{P}^{CF(k-1)}(l) \cdot \mathcal{P}^{CF(1)}(i-l)$.
- 6: **Step 3:** Obtain the probability of C-failure at each customer site i , $\mathcal{P}^{CF}(i) = \sum_{k=1}^i \mathcal{P}^{CF(k)}(i)$ ($1 \leq i \leq n_A$).

The following focuses on the calculation of the probability of the first C-failure $\mathcal{P}^{CF(1)}(i)$ ($1 \leq i \leq n_A$), which determines the probability of a C-failure at each customer site, $\mathcal{P}^{CF}(i)$ ($1 \leq i \leq n_A$).

4.1.3. Probability of the first C-failure at a customer site

In Section States when a vehicle meets the first C-failure at a customer site, the probability of vehicle state (z_i, r_i) ($i \in C^{\tau_A}$) at a customer site is used to denote the probability of the first C-failure at that customer site. Section State transition when a vehicle visits a new customer shows the state transition from (z_{l-1}, r_{l-1}) to (z_l, r_l) ($l \leq i, i \in C^{\tau_A}$); that is, how the inventory of new products z_{l-1} and available capacity for picked up items r_{l-1} are altered after a vehicle arrives at customer site l and meets the demands ξ_l and φ_l . Section Transition probabilities computes the probability of state

(z_l, r_l) at customer site l , given the probability of states (z_{l-1}, r_{l-1}) ($\forall (z_{l-1}, r_{l-1}) \in \Psi_{l-1}$) after visiting customer $l-1$. Section *Probability when a vehicle meets first C-failure at a customer site* determines the probability of the first C-failure at each customer site i ($1 \leq i \leq n_A$).

States when a vehicle meets the first C-failure at a customer site. $P^{CF(1)}(i)$ refers to the probability that vehicle V_A encounters the first C-failure at customer site i when the vehicle follows routing sequence $\tau_A = [0, 1, \dots, i-1, i, i+1, \dots, n_A, 0]$. A C-failure occurs at customer site i , so the vehicle state (z_i, r_i) satisfies $z_i \leq 0$ and $r_i \leq 0$. Further, it is the first C-failure when travelling to customer i , so the state transition for the vehicle is not interrupted by a recourse action.

Let $p_i(z_i, r_i)$ denote the probability that a vehicle travels to customer i with state (z_i, r_i) and no previous state (z_l, r_l) ($l = 1, 2, \dots, i-1$) is shown as a C-failure. Therefore, probability $P^{CF(1)}(i)$ is the sum of all probabilities $p_i(z_i = z, r_i = r)$ which show a C-failure at customer site i ; that is, any state (z_i, r_i) with $z_i = z \leq 0$ and $r_i = r \leq 0$. The relation is

$$P^{CF(1)}(i) = \sum_{z \leq 0, r \leq 0} p_i(z_i = z, r_i = r). \quad (7)$$

State transition when a vehicle visits a new customer. Assume that no C-failure occurs before visiting customer i (customer site i is the place where the first C-failure occurs). The vehicle visits a preceding customer $l-1$ ($l-1 \leq i-1$) and arrives at the next customer site l with state (z_{l-1}, r_{l-1}) . After observing demand ξ_l and φ_l of customer l , the vehicle first unloads new products from its existing inventory, resulting in vacated space Δz_l , then picks up returned items on the basis of its previous available capacity plus the newly vacated space Δz_l . After serving customer l , the inventory of new products and residual capacity are altered accordingly:

$$r_l = \begin{cases} r_{l-1} + \Delta z_l - \varphi_l, & r_{l-1} > 0, \\ \Delta z_l - \varphi_l, & r_{l-1} \leq 0. \end{cases} \quad (9)$$

In Eq. (8), the inventory of new products z_{l-1} is reduced to meet delivery demand ξ_l , either in the case of some being available on the vehicle (when $z_{l-1} > 0$) or in the case of there being no inventory to provide (when $z_{l-1} \leq 0$); unmet demand is recorded if the resulting inventory z_l takes a negative value; available capacity is temporarily increased if new products can be unloaded from the vehicle, and increment Δz_l equals $\min\{z_{l-1}, \xi_l\}$. In (9), capacity r_{l-1} after considering Δz_l is applied to satisfy demand φ_l ; the explanation of the pickup is similar to that of the delivery.

Eq. (8) shows that inventory z_l ($l \leq i$) is relevant to only the delivery demand of each customer, ξ_l ($l \leq i$). But available capacity r_l ($l \leq i$) is influenced by both the pickup and delivery demands of the customer, φ_l and ξ_l ($l \leq i$), as shown in Eq. (9), where pickup demand φ_l reduces capacity r_l directly, and delivery demand ξ_l influences capacity r_l through vacated space Δz_l . Therefore, when the vehicle travels to customer site l with state $(z_l = z, r_l = r)$, the probability $p_l(z_l = z, r_l = r)$ can be rewritten as

$$p_l(z_l = z, r_l = r) = p(z_l = z) \cdot p(r_l = r | z_l = z) = p(z_l = z) \cdot p(r_l = r | \Delta z_l), \quad \forall (z_l, r_l) \in \Psi_l. \quad (10)$$

With Eqs. (10), (8), and (9), a joint probability $p_l(z_l = z, r_l = r)$ can be obtained by first clarifying the transition of the inventory of new products from z_{l-1} to z_l , and then exploring the transition of capacity from r_{l-1} to r_l , based on vacated space Δz_l . Therefore, given $p_{l-1}(z_{l-1}, r_{l-1})$, the probability for state $(z_l = z, r_l = r)$ at customer site l , $p_l(z_l = z, r_l = r)$, is computed, as in the following section.

Transition probabilities. Given $p_{l-1}(z_{l-1}, r_{l-1})$, the probability of state (z_{l-1}, r_{l-1}) ($\forall (z_{l-1}, r_{l-1}) \in \Psi_{l-1}$), Eq. (11) computes probability $p_l(z_l = z, r_l = r)$.

$$p_l(z, r) = \begin{cases} \text{If } z > 0, r > 0, \\ \left\{ \begin{aligned} & \sum_{\substack{0 \leq e \leq Q-z \\ e < r}} q(\xi_l = e) \cdot \sum_{\substack{0 \leq t \leq Q+(e-r) \\ t \leq Q-(z+r)}} h(\varphi_l = t) \cdot p_{l-1}(z+e, r+t-e) \\ & + \sum_{\substack{0 \leq e \leq Q-z \\ e \geq r}} q(\xi_l = e) \cdot \left[\sum_{\substack{e-r \leq t \leq Q \\ t \leq Q-(z+r)}} h(\varphi_l = t) \cdot p_{l-1}(z+e, r+t-e) + h(\varphi_l = e-r) \cdot \sum_{s \leq 0} p_{l-1}(z+e, s) \right] \end{aligned} \right\} \\ \text{If } z > 0, r \leq 0, \\ \left\{ \begin{aligned} & \sum_{\substack{0 \leq e \leq Q-z \\ e \leq Q+r}} q(\xi_l = e) \cdot \left[\sum_{\substack{e-r \leq t \leq Q \\ t \leq Q-(z+r)}} h(\varphi_l = t) \cdot p_{l-1}(z+e, r+t-e) + h(\varphi_l = e-r) \cdot \sum_{s \leq 0} p_{l-1}(z+e, s) \right] \end{aligned} \right\} \\ \text{If } z \leq 0, r > 0, \\ \left\{ \begin{aligned} & \sum_{\substack{-z \leq e \leq Q \\ e < r-z}} q(\xi_l = e) \cdot \sum_{\substack{0 \leq t \leq Q+(e+z-r) \\ t \leq Q-r}} h(\varphi_l = t) \cdot p_{l-1}(z+e, r+t-(z+e))+ \\ & \sum_{\substack{-z \leq e \leq Q \\ e \geq r-z}} q(\xi_l = e) \cdot \left[\sum_{\substack{z+e-r \leq t \leq Q \\ t \leq Q-r}} h(\varphi_l = t) \cdot p_{l-1}(z+e, r+t-(z+e)) + h(\varphi_l = z+e-r) \cdot \sum_{s \leq 0} p_{l-1}(z+e, s) \right] \\ & + q(\xi_l = -z) \cdot \sum_{0 \leq t \leq Q-r} h(\varphi_l = t) \sum_{u \leq 0} p_{l-1}(u, r+t) \end{aligned} \right\} \\ \text{If } z \leq 0, r \leq 0, \\ \left\{ \begin{aligned} & \sum_{\substack{-z \leq e \leq Q \\ e \leq Q+r-z}} q(\xi_l = e) \cdot \left[\sum_{\substack{z+e-r \leq t \leq Q \\ t \leq Q-r}} h(\varphi_l = t) \cdot p_{l-1}(z+e, r+t-(z+e)) + h(\varphi_l = z+e-r) \cdot \sum_{s \leq 0} p_{l-1}(z+e, s) \right] \\ & + q(\xi_l = -z) \cdot \sum_{-r \leq t \leq Q} h(\varphi_l = t) \sum_{u \leq 0} p_{l-1}(u, r+t) \end{aligned} \right\} \end{cases} \quad (11)$$

$$z_l = \begin{cases} z_{l-1} - \xi_l, & z_{l-1} > 0, \\ -\xi_l, & z_{l-1} \leq 0. \end{cases}$$

$$\text{where, } \Delta z_l = \begin{cases} \min\{z_{l-1}, \xi_l\}, & z_{l-1} > 0, \\ 0, & z_{l-1} \leq 0. \end{cases} \quad (8)$$

Eq. (11) presents four possible vehicle states (z_l, r_l) at customer site l . These include (i) successfully serving customer l (i.e., $(z_l = z > 0, r_l = r > 0)$), (ii) failing to pick up all returned items but completing delivery to customer l (i.e., $(z_l = z > 0, r_l = r \leq 0)$), (iii) failing to satisfy delivery demand for customer l but completing its pickup (i.e., $(z_l = z \leq 0, r_l = r > 0)$), and (iv) completely failing to satisfy customer l demands for both delivery and pickup (i.e., $(z_l = z \leq 0, r_l = r \leq 0)$). Note that situations (ii) and (iii) indicate that an

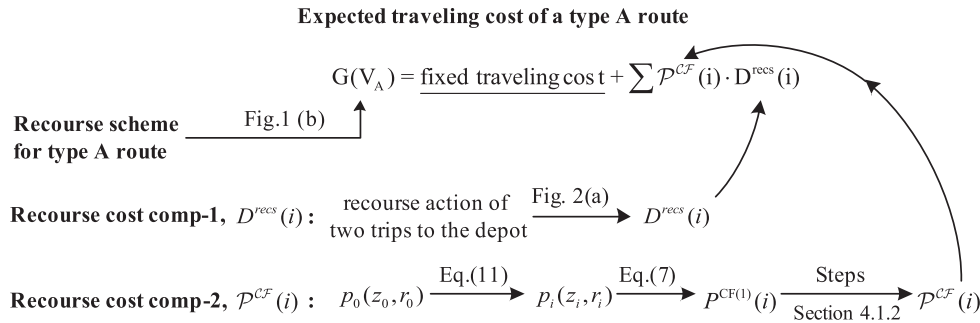


Fig. 3. Process for computing expected cost of type A route.

S-failure occurs at customer site l , and situation (iv) indicates that a C-failure occurs at customer site l .

Eq. (11) shows how state (z_{l-1}, r_{l-1}) transits to state (z_l, r_l) . In each situation, z_{l-1} takes value to ensure $z_l = z$. Then, the value r_{l-1} takes is determined given the vacated space after z_{l-1} transits to z_l . The values are obtained by reversely calculating z_{l-1} and r_{l-1} when $z_l = z$ and $r_l = r$, based on Eqs. (8) and (9). Besides, $\xi_l = e$ and $\varphi_l = t$ take values accordingly, to ensure that the transition from state (z_{l-1}, r_{l-1}) is feasible. A.1 illustrates these four situations in sequence. Further proof from another perspective is presented in A.2.

Probability when a vehicle meets first C-failure at a customer site. Eq. (11) computes probability $p_l(z_l, r_l)$ based on probability $p_{l-1}(z_{l-1}, r_{l-1})$ ($\forall (z_{l-1}, r_{l-1}) \in \Psi_{l-1}$). Therefore, given probability $p_0(z_0, r_0)$ that vehicle V_A departs from the depot with an initial inventory of new products θ_0 , probabilities $p_1(z_1, r_1), p_2(z_2, r_2), \dots, p_{i-1}(z_{i-1}, r_{i-1})$ ($\forall (z_l, r_l) \in \Psi_l, 1 \leq l \leq i-1$) can be sequentially computed, and, finally, probability $p_i(z_i, r_i)$ ($i = 1, \dots, n_A$) is obtained, in which customer site i is the place where the first C-failure occurs. The probability of initial state (z_0, r_0) is

$$p_0(z_0, r_0) = \begin{cases} 1, & \text{if } z_0 = \theta_0, r_0 = Q - \theta_0, \\ 0, & \text{else.} \end{cases} \quad (12)$$

Based on Eqs. (12) and (11), probabilities $p_1(z_1, r_1), p_2(z_2, r_2), \dots$, to $p_{n_A}(z_{n_A}, r_{n_A})$ ($\forall (z_l, r_l) \in \Psi_l, 1 \leq l \leq n_A$) are sequentially computed. Finally, the first probability at each customer site i ($1 \leq i \leq n_A$), $P^{CF(1)}(i)$ is obtained from Eq. (7).

4.1.4. Expected travelling cost of vehicle V_A

Section 4.1.3 computes the probability that vehicle V_A meets the first C-failure at a customer site; that is, $P^{CF(1)}(i)$ ($i \in C^{TA}$). Then, substituting probability $P^{CF(1)}(i)$ into Step 1 of the pseudocode in Section 4.1.2, the probability that a C-failure occurs at a customer site (i.e., $\mathcal{P}^{CF}(i)$ ($i \in C^{TA}$)) is obtained. Further, another component for obtaining the recourse cost of vehicle V_A (i.e., $D^{rec}(i)$) is computed by Eq. (1) in Section 4.1.1. Therefore, the expected cost of vehicle V_A , $G(V_A)$, is presented by

$$G(V_A) = \sum_{i=0}^{n_A-1} d(i, i+1) + d(n_A, 0) + \sum_{i=1}^{n_A-1} \mathcal{P}^{CF}(i) \cdot (3d(i, 0) + d(0, i+1) - d(i, i+1)) + \mathcal{P}^{CF}(n_A) \cdot (2d(n_A, 0)) \quad (13)$$

where $\sum_{i=0}^{n_A-1} d(i, i+1) + d(n_A, 0)$ is the length of fixed route τ_A , and $\sum_{i=1}^{n_A-1} \mathcal{P}^{CF}(i) \cdot (3d(i, 0) + d(0, i+1) - d(i, i+1)) + \mathcal{P}^{CF}(n_A) \cdot (2d(n_A, 0))$ is the expected recourse cost. Fig. 3 shows the process for obtaining the expected travelling cost of a type A route.

4.2. Expected cost of a type B route

Section 4.1 computes the expected travelling cost $G(V_A)$ of vehicle V_A which follows a type A route. Section 4.2 calculates the expected travelling cost $G(V_B)$ of vehicle V_B , which is the partner of vehicle V_A and is assigned a type B route. Vehicle V_B serves the rest of customers in subset C_p whose demands are not completely met, travelling as shown in Fig. 1(c). Section 4.2.1 gives the probability of a customer who is originally served by vehicle V_A and is revisited by vehicle V_B , as well as the possible amount of demand left unmet for the customer. Section 4.2.2 determines where to insert the customers into the a priori route, and develops an extend-and-skip approach to schedule customer visits by vehicle V_B based on an extended type B route. Section 4.2.3 presents a possible recourse strategy to reduce the total travelling cost when vehicle V_B independently meets the remaining unmet demands. With this strategy, a vehicle can take one of the two recourse actions (i.e., make two trips to the depot or a single trip to the depot) dynamically to serve customers based on a predefined routing sequence. Section 4.2.4 summarises the expected cost $G(V_B)$ of vehicle V_B under our FSC strategy.

Assume that vehicle V_B is initially assigned a type B route $\tau_B = [0, 1, \dots, j-1, j, j+1, \dots, n_B, 0]$, where j ($j \in C^{TB}$) refers to the j th customer in the routing sequence, as shown in Fig. 1(a). Let i' ($i' \in C_{unmet}^{TA}$) be the customer whose demand is unmet by vehicle V_A ($C_{unmet}^{TA} \subset C^{TA}$), where i is written i' to signify that its demand is unmet by the type A route. Each customer i' ($i' \in C_{unmet}^{TA}$) should be added to route τ_B , ensuring that all remaining demands are met by vehicle V_B .

4.2.1. Probability of adding a customer to a type B route

The possibility that vehicle V_B visits customer i ($i \in C^{TA}$) depends on the probability that an S-failure occurs at the customer site. As using FSC strategy in Fig. 1(b) shows, only customers with whose demands are not met by vehicle V_A are assigned to partner vehicle V_B . Let $P_{V_B}^{visit}(i')$ be the probability that customer i ($i \in C^{TA}$) needs to be revisited by partner vehicle V_B , and obtain this value with

$$P_{V_B}^{visit}(i') = \sum_{l=0}^{i-1} \mathcal{P}^{CF}(l) \cdot \left(\sum_{z>0, r<0} p_{i-l}(z_{i-l} = z, r_{i-l} = r) + \sum_{z<0, r>0} p_{i-l}(z_{i-l} = z, r_{i-l} = r) \right) \quad i \in C^{TA}. \quad (14)$$

Eq. (14) computes the probability that an S-failure occurs at customer site i since the last C-failure occurred at preceding customer l ($1 \leq l \leq i-1$). $\sum_{z>0, r<0} p_{i-l}(z_{i-l} = z, r_{i-l} = r) + \sum_{z<0, r>0} p_{i-l}(z_{i-l} = z, r_{i-l} = r)$ is the sum of all probabilities which show a first S-failure at customer site i since the last C-failure at the preceding customer site l , and the deduction is similar to that of Eqs. (7) and (5).

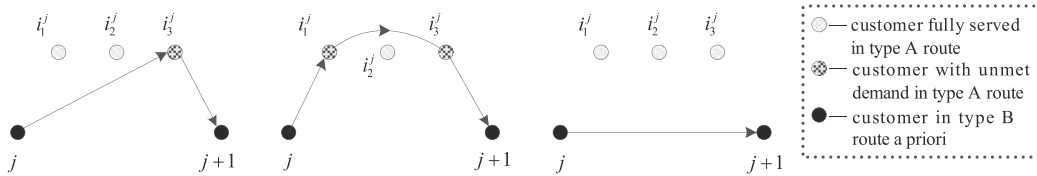


Fig. 4. Possible skips when serving all remaining customers.

Eq. (14) calculates the probability that customer i ($i \in C^{\tau_A}$) is revisited by vehicle V_B . Possible unmet demand should be either delivery demand or pickup demand for the customer, i.e., unmet delivery demand $\xi'_i = e'$ ($0 < e' \leq Q$) and completed pickup demand $\varphi'_i = 0$, or unmet pickup demand $\varphi'_i = t'$ ($0 < t' \leq Q$) and completed delivery demand $\xi'_i = 0$, when vehicle V_B revisits customer i . The probability that customer i ($i \in C^{\tau_A}$) has unmet delivery demand ξ'_i ($0 < \xi'_i \leq Q$) or unmet pickup demand φ'_i ($0 < \varphi'_i \leq Q$) is computed by

$$f(e', t') = \begin{cases} \sum_{l=0}^{i-1} p^{CF}(l) \cdot (\sum_{r=0} p_{i-l}(z_{i-l} = -e', r_{i-l} = r)), & \text{if } 0 < e' \leq Q, t' = 0, \\ \sum_{l=0}^{i-1} p^{CF}(l) \cdot (\sum_{z=0} p_{i-l}(z_{i-l} = z, r_{i-l} = -t')), & \text{if } e' = 0, 0 < t' \leq Q, \\ 0, & \text{else.} \end{cases} \quad (15)$$

The explanation of Eq. (15) is similar to the explanation of Eq. (14).

4.2.2. Extend-and-skip

Vehicle V_B meets all remaining customer demands in the shared customer set C_p . Customer j ($j \in C^{\tau_B}$), who is assigned to vehicle V_B a priori, is served by that vehicle. Other customers who initially belong to a type A route may be visited by vehicle V_B , with probability $P_{V_B}^{\text{visit}}(i')$ ($i' \in C^{\tau_A}$). The extend-and-skip approach is to extend the a priori type B route τ_B into route τ_B^{extend} , which covers all customers in set C_p ($C_p = C^{\tau_A} \cup C^{\tau_B}$), then, by skipping the customers whose demands are met during the type A route, vehicle V_B can serve all remaining customers in the shared customer set.

Route τ_B^{extend} is generated by inserting all customers i ($i \in C^{\tau_A}$) into the a priori type B route $\tau_B = [0, 1, \dots, j-1, j, j+1, \dots, n_B, 0]$, based on a simple minimum increment principle, $\min(d(l, i) + d(i, l+1) - d(l, l+1))$, where route τ_B is initially τ_B^{local} equivalent to route τ_B , and it is updated with each insertion of a new customer.

The local routing sequence of route τ_B^{extend} is $\tau_B^{\text{local-j}} = [\dots, j, i_1^j, \dots, i_m^j, j+1, \dots]$; j and $j+1$ are the customers in type B route τ_B a priori; m customers are inserted between customers j and $j+1$, and m can equal 0 to indicate there is no customer i ($i \in C^{\tau_A}$) inserted between them; i_l^j refers to the l th ($1 \leq l \leq m$) customer that is inserted after customer j . Customers j and $j+1$ are visited with 100% probability, but customer i_l^j ($1 \leq l \leq m$) is visited with probability $P_{V_B}^{\text{visit}}(i_l^j)$, or skipped with probability $1 - P_{V_B}^{\text{visit}}(i_l^j)$. In an example where three customers ($m=3$) are inserted between customers j and $j+1$, Fig. 4 shows possible skips for vehicle V_B .

In Fig. 4(a), two customers are skipped because the demands of customers i_1^j and i_2^j are met, and only the demand of customer

i_3^j is not met during the type A route. Fig. 4(b) shows the situation where only the demand of customer i_2^j is met by vehicle V_A . Fig. 4(c) shows that all three customers are skipped and vehicle V_B proceeds to directly serve its originally planned customer $j+1$. Fig. 4 only lists some of the possible scenarios for travel by vehicle V_B .

As shown in Fig. 4, in a local routing sequence $\tau_B^{\text{local-j}} = [\dots, j, i_1^j, \dots, i_m^j, j+1, \dots]$, vehicle V_B visit a customer next from current customer, depending on the probability of skipping customers between the current customer and the next customer. Let J , L , and J^{next} respectively denote the customer the vehicle is currently visiting, the customer skipped because of met demand, and the customer the vehicle will visit next. Let all possible next-visit customers, after current customer J , consist of set C_J^{next} . For example, if the vehicle visits customer J now, the next-visit customer set C_J^{next} is composed of $\{J+1, \dots, j+1\}$ and $J^{\text{next}} \in C_J^{\text{next}}$. The probability that vehicle V_B proceeds to next customer J^{next} ($J^{\text{next}} \in C_J^{\text{next}}$) from current J ($J \in \{j, i_1^j, \dots, i_m^j\}$) is determined by

$$p_{JJ^{\text{next}}} = \begin{cases} \prod_{J+1 \leq L \leq J^{\text{next}}-1} (1 - p_{V_B}^{\text{visit}}(L)) \cdot p_{V_B}^{\text{visit}}(J^{\text{next}}), & \forall J^{\text{next}} \in C_J^{\text{next}}, \text{ if } J^{\text{next}} > J+1, \\ p_{V_B}^{\text{visit}}(J^{\text{next}}), & \forall J^{\text{next}} \in C_J^{\text{next}}, \text{ if } J^{\text{next}} = J+1. \end{cases} \quad (16)$$

Note that probability $p_{JJ^{\text{next}}}$ is obtained, based on the assumption that the probability that vehicle V_B visits or skips a customer can be independently estimated by $p_{V_B}^{\text{visit}}(\cdot)$ or $1 - p_{V_B}^{\text{visit}}(\cdot)$. We approximate the probability as in Eq. (16), to facilitate the computation.

4.2.3. Recourse action for a vehicle that performs a type B route

As shown in Fig. 2(b), the recourse action of a single trip to the depot involves less travelling distance than the recourse action of two trips to the depot. Let vehicle V_B dynamically choose the recourse action that minimises the total travelling cost, when a failure occurs at a customer site. Vehicle V_B does not get any help from other vehicles; therefore, it should take a detour trip to the depot to unload and reload immediately, no matter what type of failure occurs at the customer site.

A recourse strategy, proposed by Pandelis et al. (2013), has shown the effectiveness of a vehicle following a predefined routing sequence and dynamically taking a recourse action (i.e., either one trip or two trips to the depot), when a failure occurs at a customer site. This recourse strategy formulates the routing problem as a Markov decision process (MDP), and computes the expected cost-to-go value $g_j(z_j, r_j)$ at each stage j ($j = n_B, n_B - 1, \dots, 1, 0$) and state (z_j, r_j) ($(z_j, r_j) \in \Psi_j$), for a route $\tau = [0, 1, \dots, j, j+1, \dots, n_B, 0]$. The expected cost-to-go value $g_j(z, r)$ at customer j with state $(z_j = z, r_j = r)$ is computed as

$$g_j(z, r) = \begin{cases} d(j, j+1) + \mathbb{E}g_{j+1}(z - \xi_{j+1}, r + \min(z, \xi_{j+1}) - \varphi_{j+1}), & \text{if } z \geq 0, r \geq 0 \\ 2d(j, 0) + \min\{g_j^1(z, r), g_j^2(z, r)\}, & \text{else} \end{cases} \quad (17)$$

$$\text{where } \begin{cases} g_j^1(z, r) = d(j, j+1) + \min_{\theta \leq Q - \max(z_j, r_j)} \mathbb{E}g_{j+1}(\theta - \xi_{j+1}, (Q - r_j - \theta) + \min(\theta, \xi_{j+1}) - \varphi_{j+1}) \\ g_j^2(z, r) = d(j, 0) + d(0, j+1) + \min_{\theta \leq Q} \mathbb{E}g_{j+1}(\theta - \xi_{j+1}, Q - \theta + \min(\theta, \xi_{j+1}) - \varphi_{j+1}) \end{cases}$$

Eq. (17) indicates that either the vehicle visits next customer $j+1$ directly after it fulfils the demand of current customer j ($g_j(z, r) = d(j, j+1) + \mathbb{E}g_{j+1}(z - \xi_{j+1}, r + \min(z, \xi_{j+1}) - \varphi_{j+1})$ when $(z_j = z \geq 0, r_j = r \geq 0)$) or the vehicle visits next customer $j+1$ only after detouring to the depot following recourse action at customer j . The detour to the depot can consist of either one trip or two trips. The expected total distance of the detour is $2d(j, 0) + g_j^1(z, r)$ or $2d(j, 0) + g_j^2(z, r)$, depending on which one is better (i.e., $g_j(z, r) = 2d(j, 0) + \min\{g_j^1(z, r), g_j^2(z, r)\}$).

Meanwhile, the quantity of new products to reload after a failure occurs at customer j , θ_j , is determined. For example, if it is better to take a recourse action with a single trip to the depot when a failure occurs at customer j , and the vehicle shows state $(z_j = z, r_j = r)$, then the best quantity of new products to reload is determined by $\theta_j^* = \arg \min_{\theta \leq Q - \max\{z_j^-, r_j^-\}} \mathbb{E}g_{j+1}(\theta - \xi_{j+1}, (Q - r_j^- - \theta) + \min(\theta, \xi_{j+1}) - \varphi_{j+1})$ (for details see Pandelis et al., 2013).

4.2.4. Expected travelling cost of vehicle V_B

For vehicle V_B , a recourse action, as illustrated in Section 4.2.3, is taken after each failure occurs. Section 4.2.2 shows that vehicle V_B serves customers along extended route $\tau_B^{\text{extend}} = [0, 1, \dots, J-1, J, J+1, \dots, N_p, 0]$ ($N_p = |C_p|$) and skips the customers whose demands are fully met during the type A route. Therefore, the expected travelling cost of vehicle V_B should take into consideration the possibility of skipping a customer that is assigned to a type A route a priori. Under the extend-and-skip strategy, the expected cost-to-go value of vehicle V_B that has state $(z_j = z, r_j = r)$ at customer j is updated as

$$g_j(z, r) = \sum_{j^{\text{next}} \in C_j^{\text{next}}} p_{j,j^{\text{next}}} g_{j,j^{\text{next}}}(z, r). \quad (18)$$

if $z \geq 0, r \geq 0$,

$$g_{j,j^{\text{next}}}(z, r) = d(j, j^{\text{next}}) + \mathbb{E}g_{j^{\text{next}}}(z - \xi_{j^{\text{next}}}, r + \min(z, \xi_{j^{\text{next}}}) - \varphi_{j^{\text{next}}}) \\ (z_j = z, r_j = r) \in \Psi_j. \quad (19)$$

else

$$g_{j,j^{\text{next}}}(z, r) = 2d(j, 0) + \min\{g_{j,j^{\text{next}}}^1(z, r), g_{j,j^{\text{next}}}^2(z, r)\} \\ (z_j = z, r_j = r) \in \Psi_j. \quad (20)$$

$$\begin{cases} g_{j,j^{\text{next}}}^1(z, r) = d(j, j^{\text{next}}) + \min_{\theta \leq Q - \max\{z_j^-, r_j^-\}} \mathbb{E}g_{j^{\text{next}}}(\theta - \xi_{j^{\text{next}}}, \\ (Q - r_j^- - \theta) + \min(\theta, \xi_{j^{\text{next}}}) - \varphi_{j^{\text{next}}}) \\ g_{j,j^{\text{next}}}^2(z, r) = d(j, 0) + d(0, j^{\text{next}}) + \min_{\theta \leq Q} \mathbb{E}g_{j^{\text{next}}}(\theta - \xi_{j^{\text{next}}}, \\ Q - \theta + \min(\theta, \xi_{j^{\text{next}}}) - \varphi_{j^{\text{next}}}) \end{cases}$$

In the above, $g_{j,j^{\text{next}}}(z, r)$ denotes the expected cost-to-go value where the next customer visited (after current customer j) is j^{next} ($j^{\text{next}} \in C_j^{\text{next}}$)—and with probability $p_{j,j^{\text{next}}}$ of skipping any customers between customer j and j^{next} . Therefore, the expected cost-to-go value at customer site j with state $(z_j = z, r_j = r)$ is a weighted average cost. The explanation of Eqs. (19) and (20) is similar to the explanation of Eq. (17).

So, based on the bounding condition $g_{N_p}(z, r)$,

$$g_{N_p}(z, r) = g_{N_p,0}(z, r) = d(N_p, 0) + 2d(N_p, 0) \cdot 1(z_{N_p}^- + r_{N_p}^- > 0) \\ \forall (z_{N_p} = z, r_{N_p} = r) \in \Psi_{N_p} \quad (21)$$

the expected travelling cost of vehicle V_B is computed by

$$G(V_B) = g_0 = \sum_{j \in C_0^{\text{next}}} p_{0,j^{\text{next}}} \cdot g_{0,j^{\text{next}}} = \sum_{j^{\text{next}} \in C_0^{\text{next}}} p_{0,j^{\text{next}}} \cdot (d(0, j^{\text{next}}) \\ + \min_{\theta \leq Q} \mathbb{E}g_{j^{\text{next}}}(\theta - \xi_{j^{\text{next}}}, Q - \theta + \min(\theta, \xi_{j^{\text{next}}}) - \varphi_{j^{\text{next}}})) \quad (22)$$

(refer to Pandelis et al., 2013, for details of the expected costs in Eqs. (17), (19)–(21))

5. A heuristic algorithm for our FSC strategy

For customer subset C_p ($\bigcup_p C_p = C$), the pair of vehicles V_A and V_B is assigned to follow a priori type A and B routes τ_A and τ_B , requiring that the two a priori routes cover all customers in the subset; that is, $C^{\tau_A} \cup C^{\tau_B} = C_p$ and $C^{\tau_A} \cap C^{\tau_B} = \emptyset$. According to Eq. (13) in Section 4.1 and Eq. (22) in Section 4.2, the expected total travelling cost of the vehicle pair equals $G(V_A) + G(V_B)$.

We develop an ALNS heuristic algorithm to divide all customers into several groups, ($\bigcup_p C_p = C$, $C_p \cap C_{p'} = \emptyset$), and each customer group C_p is served by either a pair of vehicles (one following a type A route, and the other following a type B route) or a single vehicle that follows a type B route. Given recourse actions taken by FSC strategy, the ALNS algorithm minimises the total travelling cost by finding the optimal a priori route for each vehicle.

Section 5.1 describes the general framework of the ALNS algorithm, which is composed of an initialisation step and several heuristics for improvement. Section 5.2 generates a priori routes based on a sweep method, Section 5.3 improves the a priori routes by several proposed removal and insertion heuristics, and Section 5.4 sets the parameters of the ALNS algorithm.

5.1. Framework of the ALNS algorithm

The ALNS algorithm was first proposed, and successfully applied, to address a pickup and delivery problem (Ropke & Pisinger, 2006). Later, it solved routing problems in stochastic contexts (e.g., Laporte, Musmanno, & Vocaturro, 2010; Lei, Laporte, & Guo, 2011). The algorithm explores the solution space by iteratively removing vertices and reinserting them into appropriate positions. More than one removal and insertion heuristics are embedded in the algorithm, and are selected based on the probability of their most improving a given route (see Ropke & Pisinger, 2006).

In our ALNS algorithm, a priori routes are initialised by a sweep method. Then, the algorithm iteratively improves these routes by selecting a removal heuristic to remove several customers from their current routes, and selecting an insertion heuristic to reinsert the customers into appropriate new routes. A pool of removal and insertion heuristics is proposed, and a pair of heuristics is selected based on scores that show their ability to improve the current solution. For each predefined number of iterations, the score of each heuristic is updated, and heuristics with higher scores have a higher probability of selection. In each iteration, the current a priori routes are altered based on the one selected removal heuristic and the one selected insertion heuristic. The new solution is accepted according to an acceptance criterion, and this optimal solution is updated if and when a new optimal solution is found. The algorithm is terminated when satisfying the stopping condition, and then returns the best solution as the optimal a priori routes for the assigned vehicles. The framework of the ALNS algorithm is generalised below.

We adapt the general ALNS algorithm to cope with the paired-vehicle case. When vehicles are paired, the removal or insertion of one vehicle's assignment can influence its partner's travelling cost. Let *move* represent the *removal* of a customer from its current route or the *insertion* of a removed customer into one of the routes. Assume that a move occurs in customer subset C_p , which is assigned to vehicles V_A and V_B currently, and the a priori routes before and after the move are (τ_A, τ_B) and (τ'_A, τ'_B) , respectively.

Algorithm 2 Overall structure of the ALNS algorithm.**initialisation:**

- 1: Generate an initial solution using the modified sweep initialisation method;
- 2: Set the initial solution as the best one, and compute the total travelling cost G^{total} ;

Iteration:

- 3: Generate a new solution by applying the chosen removal and insertion heuristics (which are selected using the roulette-wheel principle and the weight associated with each heuristic);
- 4: Determine if the new solution is to be accepted according to the acceptance criteria $G^{total'} \leq G^{total} + \delta$:
- 5: **If** it is accepted, then the current solution is accepted as the new one;
- 6: **If** it is better than the recorded optimal one, then update the best solution;
- 7: Update score σ_H of each selected heuristic H according to the adaptive search strategy
- 8: **If** ρ iterations have been executed, then recalculate and update the weights;
- 9: **If** the stopping criterion is satisfied,
- 10: Stop the algorithm;
- 11: Return the final optimal solution;
- 12: **else** return and continue;

Therefore, the cost variation ΔG for the move is

$$\Delta G = \begin{cases} \Delta G(V_A) + \Delta G(V_B), & \text{if move occurs on route } \tau_A, \\ \Delta G(V_B), & \text{if move occurs on route } \tau_B. \end{cases} \quad (23)$$

Eq. (23) indicates that the travelling cost of partner vehicle V_B is influenced when a move occurs on type A route τ_A . This can be observed from the extend-and-skip method introduced in Section 4.2.2, in which the expected travelling cost of vehicle V_B is altered, because the extended route τ_B^{extend} , which covers all customers in the subset, is changed. But when the move occurs on type B route τ_B , it only influences the travelling cost of vehicle V_B itself. Cost variation ΔG is approximated in Appendix B, which is computed based on the approximation method from (Luo, Qin, Zhang, & Lim, 2016; Yang, Mathur, & Ballou, 2000).

5.2. Initialising a priori routes

We initialise a priori routes based on a sweep method. The a priori routes generated by the sweep method will be improved in subsequent optimisation. This method clusters customers into subsets by setting an arbitrary service level. For the PD (pickup and delivery) context, we set the service level to limit the C-failure rate. Customers are initially clustered per the following procedure.

Using the sweep method, a ray extends eastward from the depot and rotates counterclockwise. By assigning the first n_A^{ini} customers to a type A route and the next n_B^{ini} customers to a type B route, route pairs are constructed sequentially. We set the initial type A route to be followed clockwise, and the initial type B route to be followed counterclockwise. This forms the a priori routes that cover all customer demands in set C. The number of customers n_A^{ini} that are initially included in a type A route is defined as

$$n_A^{ini} = \arg \max_{i \geq 1} \{P^{CF(1)}(i) | \theta_0 = \min(E(\xi), Q - E(\varphi))\}. \quad (24)$$

Eq. (24) shows that the clustering of customers stops before the location of the most probable first C-failure. As shown in Fig. 3, the probability of the first C-failure at customer i ($i \geq 1$), $P^{CF(1)}(i)$, is determined based on the initial state of the vehicle when it departs from the depot with its inventory of new products z_0 and

available capacity r_0 (i.e., $p_0(z_0, r_0)$). Further, from Eq. (12), we know that the probability of the first C-failure at customer site i , $P^{CF(1)}(i)$, is relevant to the initial inventory of new products θ_0 ($z_0 = \theta_0$). We arbitrarily set the initial inventory of new products θ_0 to $\min(E(\xi), Q - E(\varphi))$ and the quantity of reloaded new products θ_i ($i \geq 1$), if needed, also to $\min(E(\xi), Q - E(\varphi))$. This simplifies the operation of the vehicle that operates a type A route, and makes the expected cost of the type A route computationally tractable. In addition, the number of customers n_B^{ini} that are initially assigned to a type B route depends on

$$n_B^{ini} = \arg \max_{i \geq 1} \{P^{CF(1)}(i) | \theta_0 = \min(2E(\xi), 2Q - 2E(\varphi))\} - n_A^{ini}. \quad (25)$$

Eq. (25) states that the combined capacity of a pair of vehicles is $2Q$, and the initial inventory of new products is arbitrarily taken as $\min(2E(\xi), 2Q - 2E(\varphi))$. Note that the quantity of reloaded new products θ_j , if needed, varies according to the strategy as illustrated in Section 4.2.3. Hence, Eq. (25) only implies the number of customers that are covered by the a priori type B route initially, but it does not mean the quantity of reloaded new products θ_j is fixed at a specific value if a vehicle operating a type B route fails at customer j .

After being initialised, the set of a priori routes R are constructed, where R^A , R^B and R^S are the subsets of type A routes, type B routes and single routes, respectively (i.e., $R^A \cup R^B \cup R^S = R$). Note that routes in subset R^S are single type B routes, in which each vehicle performs a type B route and independently serves a customer subset. In addition, a priori type B route τ ($\tau \in R^B$) can be empty (i.e., $C^\tau = \emptyset$); in this case, the partner vehicle which follows route τ is dedicated to serving the remaining demands left by another vehicle that pairs with it. The initial routes are optimised by the following removal and insertion heuristics.

5.3. Removal and insertion heuristics

We develop two removal heuristics and two categories of insertion heuristics. They are adapted from previous heuristics (e.g., in Laporte et al., 2010; Lei et al., 2011; Ropke & Pisinger, 2006). Let $l(\tau^-)$ denote that customer l is removed from route τ and $l(\tau^+, j)$ denote that customer l is inserted into route τ before customer j . τ_A and τ_B denote type A and type B routes that are served by vehicles V_A and V_B , respectively.

5.3.1. Removal heuristics

Worst removal (WR). The WR heuristic removes possibly wrongly positioned customers from current routes. The *worst* customer is the one with the largest cost variation before and after its removal.

In the above, $\min\{|R|, n'\}$ means that we select n' routes for improvement if n' is less than the total number of current routes $|R|$.

Random removal (RR). This heuristic removes n' randomly selected customers from their current routes; it ensures diversity in the selection.

5.3.2. Insertion heuristics

Insertion heuristics are divided into two categories. Category 1 includes greedy insertion (GI) and feasibility-oriented insertion (FI), and emphasises the control of fleet size. (All routes are required to be paired if possible.) Category 2 includes modified greedy insertion (M-GI) and modified feasibility-oriented insertion (M-FI), in which cost is reduced by allowing routes to stay single instead of pairing with another single route. These two categories of insertion heuristics are developed based on extensive tests, and we find that better performance is obtained when the categories are combined to balance fleet size and cost reduction.

Algorithm 3 Heuristic: Worst removal (WR).

- 1: **For** each type A route τ_A ($\tau_A \in R^A$)
- 2: Compute the cost reduction of removing one customer i ($\forall i \in C^{\tau_A}$);
- 3: Find customer i^* whose removal most reduces cost,
- 4: $i^*(\tau_A^-) = \arg \max_{i \in C^{\tau_A}} \{\Delta G(V_A) + \Delta G(V_B)\}$;
- 5: **For** each type B route τ_B ($\tau_B \in R^B \cup R^S$)
- 6: Compute the cost change of removing one customer j from route τ_B ;
- 7: Find customer j^* whose removal causes the largest change in cost,
- 8: $j^*(\tau_B^-) = \arg \max_{j \in C^{\tau_B}} \{\Delta G(V_B)\}$;
- 9: Sort the largest cost reduction of removing the worst customer of each route τ ($\tau \in R$) decreasingly;
- 10: Select the first $\min\{|R|, n'\}$ routes and remove from them, customers i^* or j^* ;

Greedy insertion (GI). Heuristic GI greedily inserts each removed customer into its best insertion position. This position is selected by a two-step evaluation. Assume that customer l will be inserted. First, simple cost increments are calculated at each possible position of each route. (To insert the customer between customers $i-1$ and i in route τ ($\tau \in R$), the cost increment is $\Delta c(i-1, i) = d(i-1, l) + d(l, i) - d(i-1, i)$.) After that, n promising positions with the smallest increments are chosen. Finally, the cost variation ΔG is computed for each promising position, and the optimal one is the position that brings the least variation among n candidates. For each inserted customer l , the steps for greedy insertion are below. Notably, the GI heuristic has an additional choice (named pair recovery choice) for generating a new route in which l is the only customer and the new route is paired with another, already existing, single type B route.

Algorithm 4 Heuristic: Greedy insertion (GI).

- 1: Find n promising positions that are associated with the smallest increment Δc ,
- 2: **case** 1.1: if inserting into an existing route τ ,
- 3: $\Delta c(i-1, i) = d(i-1, l) + d(l, i) - d(i-1, i)$ for $\forall i \in C^\tau$,
- 4: its best position is $l(\tau^+, i^*) = \arg \min_{i \in C^\tau} \{\Delta c(i-1, i)\}$;
- 5: **case** 1.2: if generating a new route with only customer l in it to pair with a single type B route,
- 6: $\Delta c(0, 0) = 2 \times d(0, l)$;
- 7: Insert customer l in the best position, with the lowest variation ΔG among n candidates,
- 8: **case** 2.1: candidate position between i^*-1 and i^* on a type A route τ_A ($\tau_A \in R^A$),
- 9: $\Delta G = \Delta G(V_A) + \Delta G(V_B)$;
- 10: **case** 2.2: candidate position between j^*-1 and j^* on a type B route τ_B ($\tau_B \in R^B \cup R^S$),
- 11: $\Delta G = \Delta G(V_B)$;
- 12: **case** 2.3: candidate as a new type A route paired with a single type B route τ_B ($\tau_B \in R^S$),
- 13: $\Delta G = G(V_A) + \Delta G(V_B)$;

In the above, case 2.3 indicates that a new vehicle V_A is assigned to follow a new type A route, with expected travelling cost $G(V_A)$, and influences the travelling cost of another vehicle which originally serves a customer subset independently.

Feasibility-oriented insertion (FI). The FI heuristic is a pair recovery process. This heuristic identifies single routes and recovers them into pairs. Let C^{re} denote the removed customers, and R^S the set of

single routes after their removal. Initially, each single route tries to find an appropriate removed customer to be paired with, as in case 2.3 of greedy insertion. After that, $\min\{|C^{re}|, |R^S|\}$ pairs are formed. For case $|C^{re}| \neq |R^S|$, a set of removed customers $C^{re'}$ ($R^{S'} = \emptyset$) or single routes $R^{S'}$ ($C^{re'} = \emptyset$) remain. If leaving removed customers unpaired ($R^{S'} = \emptyset$), each remaining customer $l \in C^{re'}$ is inserted as in cases 2.1 and 2.2 of greedy insertion; otherwise ($C^{re'} = \emptyset$), each route $\tau_a \in R^{S'}$ pairs with its best partner $\tau_b \in R^{S'}$ until at most one route is left unpaired.

Algorithm 5 Heuristic: Feasibility-oriented insertion (FI).

- 1: Recover the best $\min\{|C^{re}|, |R^S|\}$ pairs of routes
- 2: **For** each single route $\tau \in R^S$
- 3: Compute the change in cost ΔG of pairing with each route which consists only of
- 4: inserted customer $l \in C^{re}$, as in case 2.3 of greedy insertion;
- 5: Sort all pairs in order of increased cost reduction ΔG ;
- 6: Select the best $\min\{|C^{re}|, |R^S|\}$ pairs;
- 7: **If** $|C^{re}| > |R^S|$
- 8: Insert each remaining customer $l \in C^{re'}$ into the existing route as in cases 2.1 and 2.2 of greedy insertion;
- 9: **If** $|C^{re}| < |R^S|$ and $|R^{S'}| \geq 2$, then pair the remaining single routes with each other
- 10: Choose the best partner $\tau_b \in R^{S'}$ as the partner for each route $\tau_a \in R^{S'}$ according to
- 11: $\tau_b^* = \arg \min_{\tau_b \in R^{S'}} \{\min(G^{\tau_a}(V_A) + G^{\tau_b}(V_B), G^{\tau_b}(V_A) + G^{\tau_a}(V_B))\}$;

In the above, $\tau_a \in R^{S'}$ is a type A route and $\tau_b \in R^{S'}$ is a type B route, or vice versa, depending on which total cost is better (i.e., $\min\{G^{\tau_a}(V_A) + G^{\tau_b}(V_B), G^{\tau_b}(V_A) + G^{\tau_a}(V_B)\}$).

Modified greedy insertion (M-GI). Customer l can be inserted not only into an existing route, or as part of the pair recovery choice made in a case of the original greedy insertion (GI), but as the only customer in a route that is unpaired with another single route. In heuristic M-GI, pairing, if possible, is not compulsory; a route can continue to be single and avoid pairing with another single route. In this situation, $\Delta c(0, 0) = 2 \times d(0, l)$ is computed in the first step of evaluation, and $\Delta G = G(V_B)$ with only customer l in type route τ_B is computed in the second step of evaluation.

Modified feasibility-oriented insertion (M-FI). Unlike FI, the M-FI heuristic does not involve pair recovery. Single routes can be retained, and prevented from being paired with distant routes (as described in Appendix C). In this modified heuristic, each removed customer is initialised as a single type B route in which the removed customer is the only customer in the route. For example, initialising removed customer l as a type B route; that is, $C^{\tau_B} = \{l\}$. Then, each single route can either pair with another single route or continue to be single.

5.4. Settings for the ALNS algorithm

Our main contribution is to propose a failure-specific cooperative (FSC) strategy for the PD problem under discussion. The ALNS algorithm is used here only for solution diversification. Consistent with Lei et al. (2011) and Ak and Erera (2007), the settings of the ALNS algorithm are chosen based on previous experience.

5.4.1. Adaptive search

Removal and insertion heuristics are developed based on the roulette-wheel principle. Each heuristic is associated with a weight

Table 1
The five patterns of demand distribution.

	eD-eP	hD-IP	ID-hP	ID-IP	hD-hP
$\Pr(\xi = 0)$	0.2	0	0.3	0.3	0.1
$\Pr(\xi = 1)$	0.2	0.1	0.4	0.4	0
$\Pr(\xi = 2)$	0.2	0.2	0.1	0.1	0.3
$\Pr(\xi = 3)$	0.2	0.4	0.1	0.1	0.3
$\Pr(\xi = 4)$	0.2	0.3	0.1	0.1	0.3
$E(\xi)$	2	2.9	1.3	1.3	2.7
$\Pr(\varphi = 0)$	0.2	0.3	0	0.3	0
$\Pr(\varphi = 1)$	0.2	0.5	0.2	0.5	0.2
$\Pr(\varphi = 2)$	0.2	0.2	0.3	0.2	0.3
$\Pr(\varphi = 3)$	0.2	0	0.3	0	0.3
$\Pr(\varphi = 4)$	0.2	0	0.2	0	0.2
$E(\varphi)$	2	0.9	2.5	0.9	2.5

that is updated according to the performance of the heuristic. As in previous settings of the ALNS algorithm (e.g., [Lei et al., 2011](#); [Ropke & Pisinger, 2006](#)), given $|H|$ heuristics, a heuristic H is chosen with probability $w_H / \sum_{|H|} w_H$. Its weight is updated as $w'_H = \alpha \cdot w_H + (1 - \alpha) \cdot \sigma_H / \varepsilon_H$ every ρ ($\rho = 20$) iterations, where σ_H and ε_H are the score and the number of times that heuristic H is used in ρ iterations, and α is arbitrarily set to 0.1.

Score σ_H is updated based on new solutions obtained. If a new solution is the best of all recorded solutions, then $\sigma'_H = \sigma_H + 30$; if it is acceptable and represents an improvement over the current solution, then $\sigma'_H = \sigma_H + 10$; otherwise, it is worse than the current solution but still accepted, and $\sigma'_H = \sigma_H + 6$. In the next ρ iterations, σ_H restarts from 0 and is updated in the same way.

5.4.2. Acceptance and stopping condition

As with previously used settings ([Dueck, 1993](#); [Lei et al., 2011](#)), a solution is accepted according to criteria $G^{total'} \leq G^{total} + \delta$. Here, $\delta = 0.1 \times G^{total}$, as in [Lei et al. \(2011\)](#).

The search terminates when one of three criteria is satisfied: (i) the solution is not improved over several iterations (set to three herein), (ii) a specified number of iterations has been executed (such as 100), or (iii) a preset computational time limit is violated.

6. Computational study

We implement a computational study to test the performance of FSC strategy. Customers are randomly distributed in a 1000 by 1000 square. The depot is located in the middle (500,500) or corner (0,0). We consider an instance with vehicle capacity $Q = 4$, pickup demand of each customer $\xi_i \in [0, 1, 2, 3, 4]$ ($i \in C$), and delivery demand of each customer $\varphi_i \in [0, 1, 2, 3, 4]$ ($i \in C$). Five pat-

terns of demand distribution are investigated: (i) equal uniform demand for delivery and pickup (denoted by eD-eP), (ii) high demand for delivery and low demand for pickup (hD-IP), (iii) low demand for delivery and high demand for pickup (ID-hP), (iv) low demand for both delivery and pickup (ID-IP), and (v) high demand for both delivery and pickup (hD-hP). Ten instances are generated for each possible combination of (i) customer number $|C| = 5, 10, \dots, 40$ in increments of five, (ii) depot location (500,500) or (0,0), and (iii) five different patterns of demand distribution. This yields a total of $10 \times 8 \times 2 \times 5 = 800$ instances ([Table 1](#)).

FSC is a cooperative strategy that allows customers to be served through the combined efforts of paired vehicles, and which uses risk pooling. To assess the performance of our FSC strategy, it is compared with a strategy named independent service method (ISM) that involves no cooperation between vehicles, and in which the routes are operated independently. ISM strategy initialises all routes as type B routes, and the total expected cost is the sum of the expected cost of each vehicle. We evaluate its performance in terms of real travelling costs, as shown in [Fig. 1](#). [Table 2](#) presents the advantage of FSC strategy over ISM strategy.

[Table 2](#) shows that FSC strategy clearly outperforms ISM strategy (see details at [Table D.1](#)). The associated cost reductions range from 3.8% to 33.2% for the midpoint depot, and are even greater for the corner depot, ranging from 8% to 39.2%. It also implies great potential cost reduction when the vehicle is far from the depot. This also holds true with a problem size of 5 customers; although FSC strategy exhibits a temporary decline for the hD-hP & midpoint case (with a -4.4% decline), this disadvantage disappears when the depot is at the corner, in which case it yields a 19.5% improvement.

We discover other advantages of FSC strategy, such as the number of vehicles in use and the detour frequency. [Table 3](#) shows the average number of a priori routes and detours for each individual demand distribution pattern. The 'Imp.' columns specify the reductions to the total number of routes (a priori routes plus detours); for example, in the eD-eP & midpoint case with five customers, the total number of routes for FSC and ISM respectively are 4.6 (that is, $2 + 2.6$) and 6.6 (that is, $4.4 + 2.2$), and the percentage improvement of FSC is $(6.6 - 4.6) / 6.6 = 30.3\%$.

[Table 3](#) also shows that the number of vehicles used is reduced in all cases. In addition, the number of a priori routes and detours are reduced in the first four patterns of demand distribution. In the particular case of hD-hP, ISM strategy is unable to control the size of the fleet; however, FSC strategy still performs well, with a controlled number of vehicles and only a small increase in the

Table 2
Percentage improvement in FSC over ISM (real-world cost).

# Depot	# Customer	FSC vs ISM (%)				
		eD-eP	hD-IP	ID-hP	ID-IP	hD-hP
Midpoint (500,500)	5	4.0	3.8	7.3	15.0	- 4.4
	10	13.8	10.8	13.5	20.3	8.5
	15	16.1	13.6	9.6	28.2	6.1
	20	16.6	17.6	11.6	25.2	8.4
	25	15.1	14.8	15.2	25.6	14.8
	30	19.4	18.7	18.2	31.4	16.3
	35	25.7	19.6	18.7	30.4	14.0
	40	25.5	17.4	18.8	33.2	15.2
	Ave. imp.	17.0	14.5	14.1	26.2	9.9
Corner (0,0)	5	19.0	8.0	12.2	22.5	19.5
	10	28.5	15.0	16.1	22.1	15.7
	15	27.2	20.0	20.0	29.1	18.5
	20	26.3	18.5	20.1	29.1	20.8
	25	21.7	20.7	23.0	31.3	23.2
	30	32.8	23.2	26.7	36.5	23.2
	35	30.6	21.3	25.0	33.7	23.1
	40	31.0	22.6	23.9	39.2	24.3
	Ave. imp.	27.1	18.7	20.9	30.4	21.0

Table 3
Number of a priori routes and detours in FSC vs ISM.

# Customer	# Strategy	eD-eP			hD-IP			ID-hP			ID-IP			hD-hP		
		A priori routes	Detours	Imp. (%)	A priori routes	Detours	Imp. (%)	A priori routes	Detours	Imp. (%)	A priori routes	Detours	Imp. (%)	A priori routes	Detours	Imp. (%)
5	FSC	2.0	2.6	30.3	3.2	1.0	17.6	2.2	2.1	32.8	2.0	1.1	40.4	2.0	3.0	26.6
	ISM	4.4	2.2		4.3	0.8		4.3	2.1		3.8	1.4		5.0	1.4	
10	FSC	2.4	5.3	33.0	2.8	4.4	28.7	2.0	5.1	38.3	2.4	2.7	40.0	2.2	6.0	37.9
	ISM	5.0	6.5		4.4	5.7		5.0	6.5		3.9	4.6		10.0	3.2	
15	FSC	3.1	8.3	37.0	2.5	7.4	34.9	3.1	7.9	33.7	3.2	3.3	45.0	2.5	8.8	39.9
	ISM	8.0	10.1		3.3	11.9		7.9	8.7		4.1	7.7		15.0	3.8	
20	FSC	3.6	11.4	35.6	3.7	9.9	34.3	3.4	11.5	34.1	3.8	5.8	39.6	3.7	12.0	38.2
	ISM	9.9	13.4		3.3	17.4		9.6	13.0		5.0	10.9		20.0	5.4	
25	FSC	3.4	15.2	39.4	3.8	12.7	34.8	3.8	14.6	36.1	4.8	7.2	37.5	3.4	15.5	42.2
	ISM	13.0	17.7		3.7	21.6		12.2	16.6		7.0	12.2		25.0	7.7	
30	FSC	5.1	17.6	37.8	3.8	15.5	37.7	4.8	16.8	38.3	4.2	9.2	45.7	4.8	18.4	41.9
	ISM	15.0	21.5		3.8	27.2		14.7	20.3		8.0	16.7		30.0	9.9	
35	FSC	5.6	18.7	40.9	4.7	17.9	38.6	4.9	19.9	38.8	6.6	9.9	38.7	8.6	21.0	35.7
	ISM	17.9	23.2		4.4	32.4		17.3	23.2		9.0	17.9		35.0	11.0	
40	FSC	6.2	21.8	42.3	5.4	21.4	34.0	6.4	23.2	33.8	5.6	11.9	46.0	7.2	25.3	38.8
	ISM	20.0	28.5		4.1	36.5		19.4	25.3		10.0	22.4		40.0	13.1	
	Ave. imp.	-	-	37	-	-	32.6	-	-	35.7	-	-	41.6	-	-	37.7
5	FSC	2.0	2.7	31.9	2.1	1.8	25.0	2.0	2.4	31.3	2.0	1.1	36.7	2.0	2.6	28.1
	ISM	5.0	1.9		4.1	1.1		4.7	1.7		3.3	1.6		5.0	1.4	
10	FSC	2.6	5.1	36.9	2.1	5.3	28.8	2.0	5.7	31.3	2.2	3.4	34.1	2.4	6.3	34.1
	ISM	5.0	7.2		2.1	8.3		5.0	6.2		3.4	5.1		10.0	3.2	
15	FSC	2.8	9.3	34.3	2.4	7.8	34.6	2.4	8.4	34.9	3.2	4.6	32.8	2.8	9.2	36.2
	ISM	7.8	10.6		2.4	13.2		7.9	8.7		4.0	7.6		15.0	3.8	
20	FSC	3.8	11.1	38.0	3.6	9.8	36.5	3.7	11.7	32.2	4.2	5.7	36.5	4.0	12.1	36.6
	ISM	10.0	14.0		2.7	18.4		9.9	12.8		5.0	10.6		20.0	5.4	
25	FSC	3.4	16.3	36.5	4.4	13.4	32.8	4.1	14.5	36.1	4.8	7.3	39.8	4.2	15.4	40.6
	ISM	12.9	18.1		3.3	23.2		12.5	16.6		7.0	13.1		25.0	7.7	
30	FSC	5.0	16.2	43.5	4.1	16.5	36.2	4.0	17.4	38.9	4.4	9.3	44.1	3.8	20.0	40.4
	ISM	15.0	22.5		3.7	28.6		14.5	20.5		8.0	16.5		30.0	9.9	
35	FSC	5.6	19.4	40.8	4.6	18.6	36.8	5.7	20.0	37.8	6.2	9.8	40.1	5.9	22.4	38.5
	ISM	17.7	24.5		3.5	33.2		17.3	24.0		9.0	17.7		35.0	11.0	
40	FSC	6.2	23.1	39.0	5.2	21.0	38.5	4.8	24.6	35.2	5.9	12.0	44.4	5.2	25.8	41.6
	ISM	20.0	28.0		4.0	38.6		19.5	25.9		10.0	22.2		40.0	13.1	
	Ave. imp.	-	-	37.6	-	-	33.7	-	-	34.7	-	-	38.6	-	-	36.9

number of detours. This is expected, since the failures will be more frequent—especially when each customer's pickup and delivery demand is high (hD-hP). A better strategy can balance vehicle usage and detour frequency.

Finally, we compare the computational time (see Table D.2 in Appendix D). All experiments are conducted on the Matlab platform, using an Intel Core 3.0GHz processor with 8GB of memory and a Windows operating system. As shown in Table D.2, the running time of FSC strategy does not exceed twice that of SIM strategy, and can be lower; however, running time aside, FSC strategy can generate better a priori routes and provide a cooperative strategy for the PD problem with stochastic demands.

7. Conclusions

This paper considers a simultaneous pickup and delivery problem with stochastic demands, and it introduces an FSC strategy to address recourse and cooperation during multi-vehicle routing. A paired recourse strategy is extended to the pickup and delivery problem with stochastic demands. Our FSC strategy can offer good a priori routes, and serve as a baseline for relevant research that requires fixed routing sequences.

Compared with operating vehicles independently, FSC strategy shows benefit in real travelling costs—it reduces the total cost by from 3.8% to 33.2% for a midpoint depot and by from 8% to 39.2% for a corner depot. Additionally, FSC strategy initiates fewer fixed routes and reduces recourse frequency—by 32.6% to 41.6% with a midpoint depot and by 33.7% to 38.6% with a corner depot. Further, when comparing cost savings, FSC strategy demonstrates its benefits when applied to pickup and delivery problems. Simultaneous pickup and delivery problems with stochastic demands may trigger even more failures, making a flexible recourse strategy like FSC even more valuable.

To the best of our knowledge, a paired recourse strategy (i.e., PLC) was first proposed by Ak and Erera (2007), to generate a priori routes for a multivehicle fleet, under the context of VRP with stochastic demands (VRPSD). After that, Zhu et al. (2014) proposed a paired cooperative recourse (PCR) strategy, to dynamically assign customers and reoptimise routing sequences for a pair of vehicles that coordinate to serve a set of customers with stochastic demands (Gendreau, Jabali, & Rei, 2016). This paper is the first attempt to introduce a paired recourse strategy, to provide a priori routes for a fleet of vehicles, under the context of simultaneous pickups and deliveries with stochastic demands. The computational results show the benefits in applying a paired recourse strategy for routing.

We recommend that possible case studies from the real-world applications can emerge in this field. We suggest that one possible application could be the so-called ex-van sales (see Minis & Tatarakis, 2011; Pandelis et al., 2013), in which the vehicles visit customers according to the a priori routes and the demands of each customer for new and returned products are stochastic.

Future efforts can be made to explore any benefits by considering less restrictive strategies. In our work, FSC strategy is developed based on a restricted action space, i.e., for a type A route, the vehicle makes two trips to the depot and loads a fixed quantity when a C-failure occurs at a customer site. It is interesting to investigate an improved version of the FSC strategy, if the vehicles in a pair can cooperate in a more flexible pattern. It is also interesting to see if it is worthy to adapt the FSC strategy at the expense of computational time.

We should also notice that following assumptions that are not realistic: (i) identically distributed demands for all customers, and (ii) route B starts after the completion of route A. We suggest that a more realistic model could be studied in the future, by possibly including some or all of the concerns.

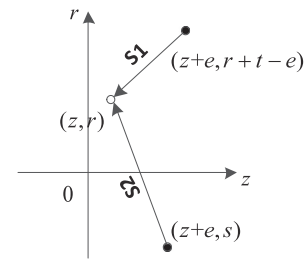


Fig. A.1. Transition to $(z_l = z > 0, r_l = r > 0)$.

Future efforts can also be made to propose a restocking policy for the problem that we concern. Preventive restocking can be performed in order to reduce the average routing cost. Minis and Tatarakis (2011) proposed a threshold-based restocking policy that manages one vehicle to replenish responsively after meeting a failure at a customer site or proactively before visiting a new customer. Future efforts can extend the work of Minis and Tatarakis (2011), by considering multiple vehicle service and including cooperation between them.

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Appendix A

Appendix A.1.

We only list situations (i) and (ii) here. Situations (iii) and (iv) are presented separately as supplementary material in the electronic companion to this paper.

A1.1. Situation (i) (when $(z_l = z > 0, r_l = r > 0)$)

In situation (i), when serving customer l successfully ($z_l > 0, r_l > 0$), state $(z_l = z, r_l = r)$ can be achieved in two cases. In case 1, the vehicle successfully serves last customer $l-1$, with sufficient inventory of new products $z_{l-1} > 0$ and enough capacity $r_{l-1} > 0$. In case 2, an S-failure occurs at last customer $l-1$, but leaves enough new products to deliver to customer l , meaning that $z_{l-1} > 0$ and $r_{l-1} \leq 0$, since vacated space Δz_l can be supplemented to capacity and may be enough to cover the pickup demand of customer l . In both cases, z_{l-1} takes $z+e$ to obtain $z_l = z_{l-1} - \xi_l = (z+e) - e = z$, and vacated space Δz_l is equal to $\Delta z_l = \min\{z+e, e\} = e$ (since $z > 0$). Moreover, in case 1, r_{l-1} should be $r+t-e$, then $r_l = r_{l-1} + \Delta z_l - \phi_l = (r+t-e) + e - t = r$ can be obtained; in case 2, for any $r_{l-1} = s \leq 0$, $r_l = \Delta z_l - \phi_l = e - (e-r) = r$ is obtained, if only pickup demand ϕ_l takes $e-r$. Fig. A.1 displays the two transitions S1 and S2, to obtain $(z_l = z > 0, r_l = r > 0)$.

In each transition, $\xi_l = e$ and $\phi_l = t$ take values to ensure the feasibility of the transition.

(a) Transition S1 refers to the circumstance that $z_{l-1} = z+e > 0$ and $r_{l-1} = r+t-e > 0$. z_{l-1} and r_{l-1} should satisfy the capacity limitation; that is, $z_{l-1} = z+e \leq Q$ and $r_{l-1} = r+t-e \leq Q$, additionally, $z_{l-1} + r_{l-1} = z+e + (r+t-e) \leq Q$. Therefore, the ranges of e and t are computed as follows.

$$\begin{aligned} \text{(i)} \quad e \text{ satisfies } & \begin{cases} 0 \leq e \leq Q \\ 0 < z+e \leq Q \end{cases} \quad (\text{since } 0 < z_{l-1} \leq Q) \Rightarrow 0 \leq e \leq Q-z. \\ \text{(ii)} \quad t \text{ satisfies } & \begin{cases} 0 \leq t \leq Q \\ 0 < r+t-e \leq Q \end{cases} \quad (\text{since } 0 < r_{l-1} \leq Q) \Rightarrow \\ & \begin{cases} 0 \leq t \leq Q \\ e-r < t \leq Q+(e-r). \end{cases} \end{aligned}$$

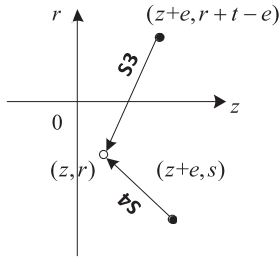


Fig. A.2. Transition to $(z_l = z > 0, r_l = r \leq 0)$.

(iii) and, $z + e + (r + t - e) \leq Q$ (since $z_{l-1} + r_{l-1} \leq Q$) $\Rightarrow t \leq Q - (z + r)$.

In (ii), $e - r < 0$ and $e - r \geq 0$ should be discussed separately: when $e - r < 0$, we obtain that,

$$\underbrace{\begin{cases} 0 \leq e \leq Q - z \\ e < r \end{cases}}_{\text{range for } e} \text{ and } \underbrace{\begin{cases} 0 \leq t \leq Q + (e - r) \\ t \leq Q - (z + r) \end{cases}}_{\text{range for } t} \quad (\text{A1})$$

(i.e., the ranges of e and t in the first term of situation $z > 0, r > 0$ in Eq. (11)) when $e - r \geq 0$, we obtain that,

$$\underbrace{\begin{cases} 0 \leq e \leq Q - z \\ e \geq r \end{cases}}_{\text{range for } e} \text{ and } \underbrace{\begin{cases} e - r < t \leq Q \\ t \leq Q - (z + r) \end{cases}}_{\text{range for } t} \quad (\text{A2})$$

(i.e., the ranges of e and t in the front part of the second term of situation $(z > 0, r > 0)$ in Eq. (11))

(b) Transition S2 refers to the circumstance that $z_{l-1} = z + e > 0$ and $r_{l-1} = \forall s \leq 0$. The capacity limitation should be satisfied, i.e., $z_{l-1} = z + e \leq Q$. Besides, as illustrated at the beginning, $r_l = r$ is obtained as long as t is $e - r$. Correspondingly, the ranges of e and t are summarised by Eq. (A3).

- (i) e satisfies $0 \leq e \leq Q - z$, as explained in Transition S1.
- (ii) t satisfies $\begin{cases} 0 \leq t \leq Q \\ t = e - r \end{cases} \Rightarrow t = e - r$ (requiring that $e - r \geq 0$).

Therefore, the transition occurs only when $e - r \geq 0$, we obtain that,

$$\underbrace{\begin{cases} 0 \leq e \leq Q - z \\ e \geq r \end{cases}}_{\text{range for } e} \text{ and } \underbrace{t = e - r}_{\text{range for } t} \quad (\text{A3})$$

(i.e., the ranges of e and t in the rear part of the second term of situation $(z > 0, r > 0)$ in Eq. (11))

$$p(z_l = z) = \begin{cases} \sum_{0 \leq e \leq Q - z} q(\xi_l = e) \cdot p(z_{l-1} = z + e) \Big|_{\Delta z_l^+ \geq 0} & z_l = z > 0, \\ \sum_{-z < e \leq Q} q(\xi_l = e) \cdot p(z_{l-1} = z + e) \Big|_{\Delta z_l^+ \geq 0} + q(\xi_l = -z) \cdot \sum_{u \leq 0} p(z_{l-1} = u) \Big|_{\Delta z_l^- = 0} & z_l = z \leq 0. \end{cases} \quad (\text{A6})$$

A1.2. Situation (ii) (when $(z_l = z > 0, r_l = r \leq 0)$)

In situation (ii), when a vehicle fails to complete pickup service of customer l , but satisfies its delivery demand ($z_l > 0, r_l \leq 0$), the state can also be achieved by the two circumstances as in situation (i); that is, transiting from state $(z_{l-1} > 0, r_{l-1} > 0)$ or from state $(z_{l-1} > 0, r_{l-1} \leq 0)$. The explanation here is similar to that of situation (i). Fig. A.2 displays the two transitions S3 and S4, to obtain $(z_l = z > 0, r_l = r \leq 0)$.

(a) Transition S3 refers to the circumstance that $z_{l-1} = z + e > 0$ and $r_{l-1} = r + t - e > 0$, mainly constrained by the capacity limitation. The explanation is similar to that in transition S1. Correspondingly, the ranges of e and t are computed as follows.

- (i) e satisfies $0 \leq e \leq Q - z$, as explained in transition S1.
- (ii) t satisfies $\begin{cases} 0 \leq t \leq Q \\ e - r < t \leq Q + (e - r) \end{cases}$, and $t \leq Q - (z + r)$, as explained in transition S1.

In (ii), $e - r \geq 0$ holds because $r \leq 0$. Additionally, transition S3 is valid only when $e - r \leq Q$. Therefore,

$$\underbrace{\begin{cases} 0 \leq e \leq Q - z \\ e \leq Q + r \end{cases}}_{\text{range for } e} \text{ and } \underbrace{\begin{cases} e - r < t \leq Q \\ t \leq Q - (z + r) \end{cases}}_{\text{range for } t} \quad (\text{A4})$$

(i.e., the ranges of e and t in the front part of situation $(z > 0, r \leq 0)$ in Eq. (11))

(b) Transition S4 refers to the circumstance that $z_{l-1} = z + e > 0$ and $r_{l-1} = \forall s \leq 0$. The explanation is similar to that in transition S2. Correspondingly, the ranges of e and t are summarised by Eq. (A5).

- (i) e satisfies $0 \leq e \leq Q - z$, as explained in transition S1.
- (ii) t satisfies $0 \leq t \leq Q$ and $t = e - r \Rightarrow t = e - r$ (requiring that $e - r \leq Q$).

Therefore, when $e - r \leq Q$, we obtain that,

$$\underbrace{\begin{cases} 0 \leq e \leq Q - z \\ e \leq Q + r \end{cases}}_{\text{range for } e} \text{ and } \underbrace{t = e - r}_{\text{range for } t} \quad (\text{A5})$$

(i.e., the ranges of e and t in the rear part of situation $(z > 0, r \leq 0)$ in Eq. (11))

Appendix A.2.

From Eq. (10), we determine $p_l(z_l = z, r_l = r)$ by obtaining probability $p(z_l = z)$ first, then finding probability $p(r_l = r | \Delta z_l)$ and distinguishing the situations where Δz_l supplements the available capacity or not. Finally, by linking $p(z_l = z) | \Delta z_l$ and $p(r_l = r | \Delta z_l)$ where Δz_l takes a different value, probability $p_l(z_l = z, r_l = r)$ can be obtained.

For distinction, the vacated space is denoted by Δz_l^+ when Δz_l can supplement capacity (i.e., $\Delta z_l^+ \geq 0$); otherwise, it is denoted by Δz_l^- when Δz_l cannot be generated from delivering new products (i.e., $\Delta z_l^- = 0$).

A2.1. $p(z_l = z)$

The probability $p(z_l = z)$ is computed by the following equation.

Based on Eq. (8), $z_l = z$ is discussed under two situations: $z_l = z > 0$ and $z_l = z \leq 0$. So $z_l = z > 0$ is derived from $z_{l-1} > 0$ when the inventory of new products is available ($\Delta z_l^+ \geq 0$), whereas $z_l \leq 0$ can be obtained from both $z_{l-1} > 0$ and $z_{l-1} \leq 0$, when the inventory is either available or not ($\Delta z_l^+ \geq 0$ or $\Delta z_l^- = 0$). The two situations are as in Eq. (A6). Moreover, from Eq. (8), $z_{l-1} = z + e$ ($e = \xi_l$) yields $z_l = z$; e should satisfy $0 < z_{l-1} = z + e \leq Q$ and $0 \leq e \leq Q$ when $\Delta z_l^+ \geq 0$ (i.e., $z_{l-1} > 0$). So, for $z_l = z > 0$, $0 \leq e \leq Q - z$, and for $z_l = z \leq 0$, $-z < e \leq Q$. Additionally, $z_l = z \leq 0$ can be obtained from any $z_{l-1} = u$ ($u \leq 0$) only when ξ_l takes $-z$.

A2.2. $p(r_l = r | \Delta z_l)$

The probability $p(r_l = r | \Delta z_l)$ is obtained by the following equation, provided that vacated space Δz_l is given.

$$p(r_l = r | \Delta z_l) = \begin{cases} \left[\sum_{0 \leq t \leq Q + (\Delta z_l - r)} h(\varphi_l = t) \cdot p(r_{l-1} = r + t - \Delta z_l) \right]_{\Delta z_l < r} & r > 0, \Delta z_l^+ \geq 0, \\ \left[\sum_{\Delta z_l - r < t \leq Q} h(\varphi_l = t) \cdot p(r_{l-1} = r + t - \Delta z_l) + h(\varphi_l = \Delta z_l - r) \cdot \sum_{s \leq 0} p(r_{l-1} = s) \right]_{\Delta z_l \geq r} & r > 0, \Delta z_l^- = 0, \\ \left[\sum_{0 \leq t \leq Q - r} h(\varphi_l = t) \cdot p(r_{l-1} = r + t) \right] & r > 0, \Delta z_l^- = 0, \\ \left[\sum_{\Delta z_l - r < t \leq Q} h(\varphi_l = t) \cdot p(r_{l-1} = r + t - \Delta z_l) + h(\varphi_l = \Delta z_l - r) \cdot \sum_{s \leq 0} p(r_{l-1} = s) \right]_{\Delta z_l \leq Q + r} & r \leq 0, \Delta z_l^+ \geq 0, \\ \left[\sum_{-r < t \leq Q} h(\varphi_l = t) \cdot p(r_{l-1} = r + t) \right] & r \leq 0, \Delta z_l^- = 0. \end{cases} \quad (A7)$$

Based on Eq. (9), $r_l = r$ is discussed under four situations: $r_l = r > 0$ and $r_l = r \leq 0$, each either with $(\Delta z_l^+ \geq 0)$ vacated space or without $(\Delta z_l^- = 0)$ vacated space. Correspondingly, $p(r_l = r > 0 | \Delta z_l^+ \geq 0)$, $p(r_l = r > 0 | \Delta z_l^- = 0)$, $p(r_l = r \leq 0 | \Delta z_l^+ \geq 0)$, and $p(r_l = r \leq 0 | \Delta z_l^- = 0)$ are obtained as in Eq. (A7).

In situation 1 ($r_l = r > 0$ and $\Delta z_l^+ \geq 0$), $r_l = r > 0$ can be obtained from both $r_{l-1} > 0$ and $r_{l-1} \leq 0$. When $r_{l-1} > 0$, from Eq. (9), $r_l = r$ is obtained from $r_{l-1} = r + t - \Delta z_l$ ($t = \varphi_l \in [0, Q]$), $0 < r_{l-1} \leq Q$, yielding $\Delta z_l - r < t \leq Q + (\Delta z_l - r)$ (with $r_{l-1} = r + t - \Delta z_l$ substituted into $0 < r_{l-1} \leq Q$). The scenarios when $\Delta z_l - r < 0$ and $\Delta z_l - r \geq 0$ should be considered separately, resulting in two terms with ranges $0 \leq t \leq Q + (\Delta z_l - r)$ and $\Delta z_l - r < t \leq Q$, respectively. For any $r_{l-1} = s$ ($s \leq 0$), from Eq. (9), $r_l = r$ can be obtained, only when φ_l equals $\Delta z_l - r$; here, $\Delta z_l - r$ must be non-negative since $\Delta z_l - r = \varphi_l \geq 0$, so $h(\varphi_l = \Delta z_l - r) \cdot \sum_{s \leq 0} p(r_{l-1} = s)$

is generalised into the scenario when $\Delta z_l \geq r$, as shown in the second term of the bottom expression when $r > 0$ and $\Delta z_l^+ \geq 0$.

In situation 2 ($r_l = r > 0$ and $\Delta z_l^- = 0$), because $z_{l-1} \leq 0$ ($\Delta z_l^- = 0$), $r_{l-1} > 0$ must be satisfied; otherwise, C-failure occurs at customer $l-1$, violating the definition of $p_l(z_l, r_l)$. In this situation, no additional capacity will be added ($\Delta z_l^- = 0$); r_{l-1} must satisfy $0 < r_{l-1} = r + t - 0 \leq Q$, and $\varphi_l = t$ should be within the range $[0, Q]$.

In situation 3 ($r_l = r \leq 0$ and $\Delta z_l^+ \geq 0$), the discussion is similar to that in situation 1, but, Δz_l^+ ($\Delta z_l^+ \geq 0$) can never be smaller than r_l ($r_l \leq 0$), so only the scenario when $\Delta z_l \geq r$ applies (i.e., with the form same as the bottom expression in situation 1); additionally, t equals $\Delta z_l - r$, which cannot exceed the limit Q , yielding $\Delta z_l \leq Q + r$.

In situation 4 ($r_l = r \leq 0$ and $\Delta z_l^- = 0$), which is similar to situation 2, t should satisfy both $0 < r_{l-1} = r + t \leq Q$ and $0 \leq t \leq Q$, so $t = \varphi_l$ is within the range $-r < t \leq Q$.

A2.3. $p_l(z_l = z, r_l = r) = p(z_l = z) \cdot p(r_l = r | \Delta z_l)$

The probability $p_l(z_l = z, r_l = r)$ is the same as in Eq. (11). $p_l(z_l = z > 0, r_l = r > 0)$ is a product of the terms in Eq. (A6) and Eq. (A7), and falls into one of four cases: $p_l(z_l = z > 0, r_l = r > 0)$, $p_l(z_l = z > 0, r_l = r \leq 0)$, $p_l(z_l = z \leq 0, r_l = r > 0)$, or $p_l(z_l = z \leq 0, r_l = r \leq 0)$.

In case 1 ($z_l = z > 0, r_l = r > 0$), $p_l(z_l = z > 0, r_l = r > 0)$ is obtained by multiplying $p(z_l = z > 0)$ in Eq. (A6) with $p(r_l = r > 0 | \Delta z_l^+ \geq 0)$ in Eq. (A7), where e replaces Δz_l in situation $r > 0$ and $\Delta z_l^+ \geq 0$ of Eq. (A7), since $z_l = z > 0$, $\Delta z_l = \min(z + e, e) = e$. Additionally, $t \leq Q - (z + r)$ —since $z_{l-1} + r_{l-1} \leq Q$ when both $z_{l-1} = z + e > 0$ and $r_{l-1} = r + t - e > 0$.

In case 2 ($z_l = z > 0, r_l = r \leq 0$), $p_l(z_l = z > 0, r_l = r \leq 0)$ equals $p(z_l = z > 0) \cdot p(r_l = r \leq 0 | \Delta z_l^+ \geq 0)$. e replaces Δz_l and $t \leq Q - (z + r)$ for the same reason as in case 1.

In case 3 ($z_l = z \leq 0, r_l = r > 0$), $p_l(z_l = z \leq 0, r_l = r > 0) = p(z_l = z \leq 0 | \Delta z_l^+ \geq 0) \cdot p(r_l = r > 0 | \Delta z_l^+ \geq 0) + p(z_l = z \leq 0 | \Delta z_l^- = 0) \cdot p(r_l = r > 0 | \Delta z_l^- = 0)$; here,

$p(z_l = z \leq 0 | \Delta z_l^+ \geq 0)$ and $p(z_l = z \leq 0 | \Delta z_l^- = 0)$ refer to two terms of the bottom expression for $p(z_l = z)$ when $z \leq 0$ in Eq. (A6). $z + e$ is substituted for Δz_l since $\Delta z_l = \min(z + e, e) = z + e$ when $z \leq 0$. $t \leq Q - r$ is obtained by applying $z_{l-1} + r_{l-1} \leq Q$ when both $z_{l-1} = z + e > 0$ and $r_{l-1} = r + t - (z + e) > 0$.

In case 4 ($z_l = z \leq 0, r_l = r \leq 0$), as in case 3, $p_l(z_l = z \leq 0, r_l = r \leq 0) = p(z_l = z \leq 0 | \Delta z_l^+ \geq 0) \cdot p(r_l = r \leq 0 | \Delta z_l^+ \geq 0) + p(z_l = z \leq 0 | \Delta z_l^- = 0) \cdot p(r_l = r \leq 0 | \Delta z_l^- = 0)$. The explanation is similar.

Appendix B.

For each move occurring on route τ ($\tau \in R$), we approximate the cost variation ΔG . We take insertion as the example (removal is similar). Eq. (B1) computes the approximated cost $\Delta G'$ when inserting customer l between customer $i^* - 1$ and i^* on a priori type A route τ_A or between customers j^* and $j^* + 1$ on a priori type B route τ_B .

$$\Delta G' = \begin{cases} \Delta G^{\tau_A}(\tau_A)(V_A) + \Delta G^{\tau_B}(\tau_B^{extend})(V_B), & \text{if move occurs on route } \tau_A, \\ \Delta G^{\tau_B}(\tau_B^{extend})(V_B), & \text{if move occurs on route } \tau_B. \end{cases} \quad (B1)$$

In the above, $\Delta G(V_A)$ and $\Delta G(V_B)$ are approximated by $\Delta G^{\tau_A}(\tau_A)(V_A)$ and $\Delta G^{\tau_B}(\tau_B^{extend})(V_B)$, respectively. We approximate the cost variation by adopting the idea from Yang et al. (2000) and Luo et al. (2016). In other words, by computing the average cost difference of inserting customer l immediately before customer i^* for a type A route, ignoring the influence on the customers after i^* (to obtain variation $\Delta G^{\tau_A}(\tau_A)(V_A)$), computing the average cost difference of inserting customer l immediately after customer j^* for a type B route, and ignoring the influence on the customers before j^* (to obtain variation $\Delta G^{\tau_B}(\tau_B^{extend})(V_B)$). By doing this, the same level of cost variation is detected and computational effort is greatly saved.

Herein, $\mathcal{P}^{i^*}(\tau_A) = \{0, 1, \dots, i^* - 1, i^*\}$ denotes a partial route of τ_A . $\Delta G^{\tau_A}(\tau_A)(V_A)$ is the difference of expected costs for routes $\{0, 1, \dots, i^* - 1, l, i^*\}$ and $\{0, 1, \dots, i^* - 1, i^*\}$, and the expected cost is computed by Eq. (13). The expression $\mathbb{P}^{j^*}(\tau_B^{extend})$ denotes a partial route of τ_B , $\mathbb{P}^{j^*}(\tau_B^{extend}) = \{j^*, i'_1, \dots, i'_m, j^* + 1, \dots, n_B\}$. Note that partner vehicle V_B actually performs extended route τ_B^{extend} , in which several customers from the a priori type A route could possibly be revisited (such as customers $i'_1, \dots, i'_m$ between customers j^* and $j^* + 1$). The expression $\Delta G^{\tau_B}(\tau_B^{extend})(V_B)$ is the cost difference between routes $\{j^*, i'_1, \dots, l, \dots, i'_m, j^* + 1, \dots, n_B\}$ and $\{j^*, i'_1, \dots, i'_m, j^* + 1, \dots, n_B\}$, and equals $\frac{\sum_{r=Q-Q}^{Q-Q} \sum_{z=Q-Q}^{Q-Q} (g'_j(z, r) - g_j(z, r))}{(2Q+1)^2 - \frac{1}{2}Q(Q+1)}$. In $\frac{\sum_{r=Q-Q}^{Q-Q} \sum_{z=Q-Q}^{Q-Q} (g'_j(z, r) - g_j(z, r))}{(2Q+1)^2 - \frac{1}{2}Q(Q+1)}$ (when inserting customer l before and

after, with vehicle state ($z_j = z, r_j = r$), $g'_j(z, r)$ and $g_j(z, r)$ are the expected cost-to-go values obtained by Eq. (18); the formula indicates the average cost of different vehicle states (z_j, r_j) ($(z_j, r_j) \in \Psi_j$).

Appendix D.

Tables D.1 and D.2.

Appendix C.

Algorithm 6 Heuristic: Modified feasibility-oriented insertion (M-FI).

- 1: initialise each removed customer as a single type B route,
- 2: $\tau_B = \{0, l, 0\}$ for each removed customer l ;
- 3: Identify all single routes and all paired routes
- 4: R^S represents the set of all single routes,
- 5: R^P is the set of all paired routes;
- 6: **For** each single route $\tau \in R^S$, find its best partner or keep it single,
- 7: Compare between being a partner for another single route and being single
- 8: Compute the cost reduction Δs when coupling with a single route $\tau' \in R^S$;
- 9: **For** type B routes τ and τ' , Δs equals,
- 10: $G^\tau(V_B) + G^{\tau'}(V_B) - \min\{G^\tau(V_A) + G^{\tau'}(V_B), G^{\tau'}(V_A) + G^\tau(V_B)\}$;
- 11: Select a candidate route with the largest positive cost reduction $\tilde{\tau} = \arg \max_{\tau' \in R^S} \{\Delta s > 0\}$;
- 12: Compare between inserting in a route pair and being single
- 13: Compute the cost reduction $\Delta s'$ when inserting into one of the paired routes;
- 14: **For** inserting on one of the routes in pair (τ_A, τ_B) , $\Delta s'$ equals,
- 15: $G^\tau(V_B) + (G^{\tau_A}(V_A) + G^{\tau_B}(V_B)) - \min\{G^{\tau_A}(V_A) + G^{\tau_B}(V_B), G^{\tau_B}(V_A) + G^{\tau_A}(V_B)\}$;
- 16: (τ'_A or τ'_B are the routes after inserting all customers of route τ into τ_A or τ_B)
- 17: Select a candidate route with the largest positive cost reduction $\tilde{\tau} = \arg \max_{\tau' \text{ in } (\tau_A, \tau_B) \in R^P} \{\Delta s' > 0\}$.
- 18: Determine to be a partner or stay single
- 19: **If** ($\forall \Delta s \leq 0$ and $\Delta s' \leq 0$),
- 20: route τ stays single, $R^S = R^S \setminus \tau$,
- 21: **Else** choose the best partner route $\tau^* = \arg \max_{\tilde{\tau}, \tilde{\tau}'} \{\Delta s > 0, \Delta s' > 0\}$, $R^S = R^S \setminus (\tau \cup \tau^*)$.

Table D1

Cost comparison of FSC vs ISM.

# Depot	# Customer	eD-eP		hD-IP		ID-hP		ID-IP		hD-hP	
		FSC	ISM	FSC	ISM	FSC	ISM	FSC	ISM	FSC	ISM
Midpoint (500,500)	5	4828.4	5031.8	4141.4	4306.9	5024.5	5421.2	4032.7	4744.3	4682.9	4486.1
	10	8525.6	9894.4	8756.0	9821.1	8863.3	10243.4	6593.9	8270.8	9328.1	10197.3
	15	13596.7	16197.5	12438.5	14394.3	13302.6	14721.2	8086.6	11263.2	13997.0	14900.8
	20	16385.6	19653.5	16113.7	19563.6	17257.5	19532.4	12077.9	16153.3	17608.7	19215.9
	25	21608.3	25444.3	20441.0	23978.4	21629.8	25510.2	14008.7	18841.1	20987.0	24634.6
	30	24928.0	30921.0	23045.3	28329.8	24642.9	30122.9	16461.0	23990.0	25414.5	30376.2
	35	26935.9	36240.2	27380.6	34063.5	27240.2	33499.3	17561.2	25228.9	30072.1	34945.7
	40	31687.6	42517.6	30442.7	36846.8	31093.9	38301.7	20610.5	30839.1	34137.5	40252.8
	5	8523.8	10531.8	7716.2	8381.4	8429.5	9604.5	6604.2	8522.3	8602.8	10687.2
	10	14476.3	20260.5	16240.0	19111.1	15241.6	18173.4	11570.0	14850.3	16542.3	19614.0
Corner (0,0)	15	23734.0	32590.1	22250.1	27792.6	21636.9	27014.7	14964.8	21109.0	23563.1	28901.4
	20	28636.5	38868.8	28120.7	34491.5	29594.8	37060.4	20122.3	28381.1	31341.7	39573.5
	25	39377.2	50268.8	35535.7	44825.2	37325.7	48474.0	24485.5	35638.4	37188.8	48451.4
	30	40439.7	60150.3	42537.8	55364.4	43174.1	58918.3	28289.8	44558.3	45748.6	59605.5
	35	48427.6	69778.6	47980.1	61003.8	51050.2	67986.8	32210.0	48567.2	54350.0	70641.1
	40	55047.2	79757.6	55254.3	71431.0	57183.8	75129.0	36038.9	59249.2	61901.5	81823.2

Table D2

Average computation time for ALNS algorithm per instance of FSC and ISM (seconds).

# Customer	Midpoint				Corner			
	FSC		ISM		FSC		ISM	
	Mean	Max	Mean	Max	Mean	Max	Mean	Max
5	10.8	14.4	11.2	15.7	4.0	9.2	4.6	10.1
10	58.0	81.1	111.3	280.0	48.4	81.7	74.8	115.3
15	218.5	249.7	166.5	193.5	136.4	215.7	133.4	202.8
20	341.7	442.6	298.5	360.8	188.9	282.4	215.5	366.4
25	578.0	772.6	424.2	523.0	197.7	331.0	213.2	359.7
30	760.1	990.9	620.5	779.6	232.4	405.3	303.1	500.1
35	671.9	877.5	863.2	1031.0	308.6	559.6	438.0	792.9
40	1568.1	2618.8	893.8	1387.6	535.2	1045.1	405.6	719.2

Table E1

Main notations.

ijl	Customers
ξ_i, φ_i	Delivery demand and pickup demand of customer i
e, t	Realised amounts for random variables ξ_i and φ_i , respectively
$q(\xi_i = e), h(\varphi_i = t)$	Probabilities when variables ξ_i and φ_i take e and t , respectively
(z_i, r_i)	States (inventory of new products and available capacity for picking up, after the first visit of customer i)
z_i^-, r_i^-	Amount of new products that cannot be delivered to customer i , and amount of returned items of customer i that cannot be loaded
Δz_i	Vacated capacity after the vehicle first unloads new products to customer i
z, u	Realised values that variable z_i takes
r, s	Realised values that variable r_i takes
k	Times of C-failure (i refers to the i th customer of the routing sequence in this sub-column)
$P^{CR(k)}(i)$	Probability that a vehicle meets the k th C-failure at customer site i
$p^{CF}(i)$	Probability of the C-failure at customer i
$p_i(z_i, r_i)$	Probability that a vehicle encounters the first C-failure at customer site i and the state shows as (z_i, r_i)

Appendix E.

Table E.1

Supplementary material

Supplementary material associated with this article can be found, in the online version, at [10.1016/j.ejor.2018.05.049](https://doi.org/10.1016/j.ejor.2018.05.049).

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